Notes on by-mode ECSW for LSPG reduced-order models

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BM-ECSW was proposed by Sebastian Rodriguez [9]. Its mode-dependent cubature weights are applied here to LSPG reduced-order models.

1 Projection-based reduced-order models

The construction of a projection-based reduced-order model (PROM) begins by collecting N_s solution snapshots $\boldsymbol{u}_s \in \mathbb{R}^N$ of a high-dimensional model (HDM), where N denotes the number of degrees of freedom of the original model. These snapshots sample the evolution of the solution at selected times and parameter values. Given a fixed reference state $\boldsymbol{u}_{\text{ref}}$, their deviations from this state are assembled into a snapshot matrix and compressed using a thin singular value decomposition (SVD):

$$\begin{aligned} \boldsymbol{S} &= [\boldsymbol{u}_1 - \boldsymbol{u}_{\text{ref}}, \dots, \boldsymbol{u}_{N_s} - \boldsymbol{u}_{\text{ref}}] \in \mathbb{R}^{N \times N_s}, \\ \boldsymbol{S} &= \boldsymbol{U}_S \boldsymbol{\Sigma}_S \boldsymbol{Y}_S^T. \end{aligned} \tag{1}$$

Here $\boldsymbol{U}_S$ and $\boldsymbol{Y}_S$ contain the left and right singular vectors, respectively, and $\boldsymbol{\Sigma}_S$ contains the singular values $\sigma_{S,i}$ in descending order. The proper orthogonal decomposition (POD) basis is formed by retaining the first $n \ll N$ left singular vectors, yielding the orthonormal reduced-order basis (ROB)

$$\boldsymbol{V} = \boldsymbol{U}_S(:, 1:n) = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{R}^{N \times n}, \qquad \boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}_n. \tag{2}$$

The reduced dimension n is selected using the singular value energy truncation criterion

$$1 - \frac{\sum_{i=1}^{n} \sigma_{S,i}^2}{\sum_{j=1}^{\min(N,N_s)} \sigma_{S,j}^2} \leq \varepsilon_S^2, \tag{3}$$

where $\varepsilon_S > 0$ controls the discarded fraction of the sum of squared singular values.

Using the ROB in Equation (2), the HDM state is approximated as

$$\widetilde{\boldsymbol{u}}(\boldsymbol{q}) = \boldsymbol{u}_{\text{ref}} + \boldsymbol{V}\boldsymbol{q} = \boldsymbol{u}_{\text{ref}} + \sum_{i=1}^{n} q_i \boldsymbol{v}_i. \tag{4}$$

In this approximation, the columns $\boldsymbol{v}_i$ of $\boldsymbol{V}$ are the POD modes, and $\boldsymbol{q} \in \mathbb{R}^n$ is the vector of generalized coordinates. The reference state and ROB remain fixed, while the coordinates depend on time and parameters. Equation (4) therefore restricts the approximate state to an n-dimensional affine trial space. The evolution of its coordinates is determined by projecting the governing equations onto a suitable test space.

1.1 Least-squares Petrov–Galerkin projection

Consider a parametric, nonlinear, semi-discrete first-order initial-value problem of the form

$$\boldsymbol{M}(\boldsymbol{\mu})\dot{\boldsymbol{u}}(t;\boldsymbol{\mu}) + \boldsymbol{f}(\boldsymbol{u}(t;\boldsymbol{\mu});\boldsymbol{\mu}) - \boldsymbol{g}(t;\boldsymbol{\mu}) = \boldsymbol{0}, \qquad \boldsymbol{u}(0;\boldsymbol{\mu}) = \boldsymbol{u}_0(\boldsymbol{\mu}). \tag{5}$$

Here $\boldsymbol{u}(t;\boldsymbol{\mu}) \in \mathbb{R}^N$ is the HDM state, $\boldsymbol{M}$ is the mass matrix, $\boldsymbol{f}$ contains the internal or flux terms, and $\boldsymbol{g}$ contains the forcing terms. The vector $\boldsymbol{\mu} \in \mathcal{D}$ specifies the parameters, and a dot denotes differentiation with respect to time. Applying an implicit time discretization to Equation (5) gives the nonlinear system

$$\boldsymbol{r}^{m+1}(\boldsymbol{u}^{m+1};\boldsymbol{\mu}) = \boldsymbol{0}, \tag{6}$$

where $\boldsymbol{r}^{m+1} \in \mathbb{R}^N$ is the time-discrete HDM residual and the superscript $m+1$ identifies the time level t^{m+1}. The parameters and all history states required by the time integrator are held fixed during the solution at this time level; the history states are omitted from the residual arguments.

Substituting the affine approximation in Equation (4) into Equation (6) yields N equations for the n generalized coordinates. This system is generally overdetermined. Least-squares Petrov–Galerkin (LSPG) projection determines the coordinates by minimizing the Euclidean norm of the time-discrete residual [2, 1]:

$$\boldsymbol{q}^{m+1} \in \operatorname*{arg\,min}_{\boldsymbol{q}\in\mathbb{R}^n} \Phi(\boldsymbol{q}), \qquad \Phi(\boldsymbol{q}) = \frac{1}{2}\left\|\boldsymbol{r}^{m+1}(\boldsymbol{u}_{\mathrm{ref}} + \boldsymbol{V}\boldsymbol{q})\right\|_2^2. \tag{7}$$

The residual in this objective is evaluated with the known approximate history states and the prescribed parameters, which are omitted from its arguments. To express the stationarity conditions of Equation (7), define the HDM residual Jacobian and the LSPG test basis as

$$\begin{aligned} \boldsymbol{J}(\widetilde{\boldsymbol{u}}) &= \frac{\partial \boldsymbol{r}^{m+1}}{\partial \widetilde{\boldsymbol{u}}} \in \mathbb{R}^{N\times N}, \\ \boldsymbol{W}(\boldsymbol{q}) &= \frac{\partial \boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q}))}{\partial \boldsymbol{q}} = \boldsymbol{J}(\widetilde{\boldsymbol{u}}(\boldsymbol{q}))\boldsymbol{V} \in \mathbb{R}^{N\times n}. \end{aligned} \tag{8}$$

The last equality follows from the chain rule and the fixed affine approximation, for which $\partial\widetilde{\boldsymbol{u}}/\partial\boldsymbol{q} = \boldsymbol{V}$. Thus the test basis depends on the current state through the HDM residual Jacobian. Differentiating the least-squares objective using Equation (8) gives the projected residual and the resulting stationarity equations:

$$\begin{aligned} \boldsymbol{r}_n(\boldsymbol{q}) &= \nabla_{\boldsymbol{q}}\Phi(\boldsymbol{q}) = \boldsymbol{W}(\boldsymbol{q})^T\boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q})), \\ \boldsymbol{r}_n(\boldsymbol{q}^{m+1}) &= \boldsymbol{0}. \end{aligned} \tag{9}$$

These n nonlinear equations express the orthogonality of the discrete HDM residual to the test space spanned by the columns of $\boldsymbol{W}$ and provide the first-order stationarity conditions for the minimization problem in Equation (7).

The nonlinear least-squares problem in Equation (7) is solved using a Gauss–Newton iteration [3]. At iteration k within time step $m+1$, let $\boldsymbol{W}^{m+1,k} = \boldsymbol{W}(\boldsymbol{q}^{m+1,k})$. Linearizing the residual with respect to a coordinate increment $\Delta\boldsymbol{q}$ around the current iterate gives

$$\boldsymbol{r}^{m+1}\big(\widetilde{\boldsymbol{u}}(\boldsymbol{q}^{m+1,k}+\Delta\boldsymbol{q})\big) \approx \boldsymbol{r}^{m+1}\big(\widetilde{\boldsymbol{u}}(\boldsymbol{q}^{m+1,k})\big) + \boldsymbol{W}^{m+1,k}\Delta\boldsymbol{q}. \tag{10}$$

The Gauss–Newton correction minimizes the Euclidean norm of this linearized residual. Its normal equations are

$$\left[\boldsymbol{W}^{m+1,k}\right]^T \boldsymbol{W}^{m+1,k}\Delta\boldsymbol{q}^{m+1,k} = -\left[\boldsymbol{W}^{m+1,k}\right]^T \boldsymbol{r}^{m+1}\big(\widetilde{\boldsymbol{u}}(\boldsymbol{q}^{m+1,k})\big), \tag{11}$$

with all quantities evaluated at the current iterate. Solving this $n\times n$ system gives the coordinate correction used in the update

$$\boldsymbol{q}^{m+1,k+1} = \boldsymbol{q}^{m+1,k} + \Delta\boldsymbol{q}^{m+1,k}. \tag{12}$$

The residual and test basis are then evaluated at the updated state to form the next correction. This procedure is repeated until the projected equations in Equation (9) satisfy the prescribed nonlinear convergence criterion.

2 Classical energy-conserving sampling and weighting (ECSW)

Although LSPG projection reduces the number of unknowns from N to $n \ll N$, assembling the reduced equations still requires evaluating the HDM residual and the LSPG test basis on the original mesh. For nonlinear PROMs, these quantities depend on the current reduced state and are evaluated at each Gauss–Newton iteration to form the system in Equation (11). The computational cost of this assembly therefore remains dependent on the HDM dimension N, which can limit the acceleration achieved in applications requiring repeated or rapid predictions. Hyperreduction addresses this dependence by evaluating only a subset of the contributions needed to assemble the reduced equations. The energy-conserving sampling and weighting (ECSW) method [4, 5, 6, 7] constructs such an approximation using positive cubature weights on a sampled mesh. Originally developed for nonlinear structural dynamics, the method has been extended to Petrov–Galerkin PROMs of first-order dynamical systems.

To construct this approximation, consider the set of mesh entities $\mathcal{E} = \{e_1, \dots, e_{N_e}\}$ associated with the HDM discretization. An entity can be a node or a cell, depending on the spatial discretization. Each entity e is associated with a block of d_e assembled residual entries, and these blocks form a nonoverlapping partition of the global residual. The Boolean matrix $\boldsymbol{L}_e \in \{0,1\}^{d_e\times N}$ selects the residual rows belonging to this block, while $\boldsymbol{L}_{e^+} \in \{0,1\}^{d_e^+\times N}$ selects the state entries required to evaluate it, including those of neighboring entities within the numerical stencil. The residual partition satisfies

$$\sum_{e\in\mathcal{E}} \boldsymbol{L}_e^T\boldsymbol{L}_e = \boldsymbol{I}_N. \tag{13}$$

This partition is applied to assembled residual rows; element contributions that share degrees of freedom are combined before forming these blocks. At a fixed time level, the residual block of entity e can be written as

$$\boldsymbol{r}_e(\boldsymbol{q}) = \boldsymbol{L}_e\boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q})) = \boldsymbol{r}_e^{m+1}(\boldsymbol{L}_{e^+}[\boldsymbol{u}_{\text{ref}} + \boldsymbol{V}\boldsymbol{q}]) \in \mathbb{R}^{d_e}, \tag{14}$$

where the prescribed parameters and known history states are held fixed and omitted from the arguments. Embedding these local blocks into the global residual space gives

$$\boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q})) = \sum_{e\in\mathcal{E}} \boldsymbol{L}_e^T \boldsymbol{r}_e(\boldsymbol{q}). \tag{15}$$

Substituting this assembly into Equation (9) expresses the LSPG projected residual as a sum over the original mesh:

$$\begin{aligned} \boldsymbol{r}_n(\boldsymbol{q}) &= \boldsymbol{W}(\boldsymbol{q})^T \boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q})) \\ &= \sum_{e\in\mathcal{E}} \boldsymbol{W}(\boldsymbol{q})^T \boldsymbol{L}_e^T \boldsymbol{r}_e(\boldsymbol{q}). \end{aligned} \tag{16}$$

The contribution of each entity is obtained by restricting the test basis to its residual rows. Defining

$$\boldsymbol{W}_e = \boldsymbol{L}_e \boldsymbol{W} = \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}, \qquad \boldsymbol{g}_e = \boldsymbol{W}_e^T \boldsymbol{r}_e, \qquad \boldsymbol{r}_n = \sum_{e\in\mathcal{E}} \boldsymbol{g}_e, \tag{17}$$

gives $\boldsymbol{W}_e \in \mathbb{R}^{d_e \times n}$ and $\boldsymbol{g}_e \in \mathbb{R}^n$. Thus each local residual block supplies a contribution to all n reduced equations.

ECSW approximates the full projected sum in Equation (16) using a sampled subset $\widetilde{\mathcal{E}} \subset \mathcal{E}$, with $n_e = |\widetilde{\mathcal{E}}| \ll N_e$, and one positive scalar weight ξ_e per selected entity:

$$\begin{aligned} \boldsymbol{r}_n(\boldsymbol{q}) &\approx \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}) \\ &= \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}(\boldsymbol{q})^T \boldsymbol{L}_e^T \boldsymbol{r}_e(\boldsymbol{q}) = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{g}_e(\boldsymbol{q}). \end{aligned} \tag{18}$$

The weight ξ_e multiplies every component of $\boldsymbol{g}_e$, so it is shared by all reduced equations. The sampled mesh and these weights are trained together so that the weighted contributions approximate the full projected residual over a set of training configurations to within a prescribed tolerance [7]. Once this cubature rule is fixed, the generalized coordinates are obtained at each time step by setting the approximated projected residual to zero.

Evaluating the sampled sum in Equation (18) requires the state entries selected by $\boldsymbol{L}_{e^+}$ for every $e \in \widetilde{\mathcal{E}}$. These dependencies can extend beyond the sampled entities. The union of the sampled mesh and its required stencil neighbors defines the augmented mesh $\widetilde{\mathcal{E}}^+$, with

$$n_e = |\widetilde{\mathcal{E}}|, \qquad n_e^+ = |\widetilde{\mathcal{E}}^+|, \qquad n_e \le n_e^+ \le N_e. \tag{19}$$

Both sizes count mesh entities. Let $\mathcal{I}^+$ denote the scalar state indices belonging to $\widetilde{\mathcal{E}}^+$. Restricting the reference state and the ROB to these indices gives

$$\boldsymbol{u}_{\mathrm{ref}}^+ = \boldsymbol{u}_{\mathrm{ref}}(\mathcal{I}^+), \qquad \boldsymbol{V}^+ = \boldsymbol{V}(\mathcal{I}^+, :). \tag{20}$$

The affine approximation can then be evaluated on the augmented mesh as

$$\widetilde{\boldsymbol{u}}^+(\boldsymbol{q}) = \boldsymbol{u}_{\mathrm{ref}}^+ + \boldsymbol{V}^+ \boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}^+|}. \tag{21}$$

Using this restricted state to evaluate the selected residual blocks, the corresponding test-basis blocks follow from the chain rule:

$$\boldsymbol{W}_e(\boldsymbol{q}) = \frac{\partial \boldsymbol{r}_e}{\partial \widetilde{\boldsymbol{u}}^+} \boldsymbol{V}^+, \qquad e \in \widetilde{\mathcal{E}}. \tag{22}$$

Here the local residual derivative has dimensions $d_e \times |\mathcal{I}^+|$ and $\boldsymbol{V}^+$ has dimensions $|\mathcal{I}^+| \times n$. Consequently, the state reconstruction and local evaluations needed during the online iterations depend on the reduced dimension and the sizes of the sampled and augmented meshes.

2.1 Offline mesh sampling and weight fitting

The cubature rule in Equation (18) is constructed from N_c training configurations, each comprising a trial state, its parameters, and the known history required by the time-discrete residual. These states can be obtained from the HDM solution snapshots by orthogonal projection onto the affine trial space:

$$\overline{\boldsymbol{u}} = \boldsymbol{u}_{\mathrm{ref}} + \boldsymbol{V}\boldsymbol{V}^T[\boldsymbol{u}_{\mathrm{HDM}} - \boldsymbol{u}_{\mathrm{ref}}]. \tag{23}$$

When the residual depends on history states, the same projection is applied to those snapshots. Evaluating the local residuals and test-basis blocks at each configuration $c = 1, \ldots, N_c$ gives the projected entity contributions and their full sum:

$$\boldsymbol{g}_e^{(c)} = [\boldsymbol{W}_e^{(c)}]^T \boldsymbol{r}_e^{(c)} \in \mathbb{R}^n, \qquad \boldsymbol{b}^{(c)} = \sum_{e \in \mathcal{E}} \boldsymbol{g}_e^{(c)}. \tag{24}$$

Stacking these contributions over the training configurations produces the matrix and target vector

$$\boldsymbol{G} = \begin{bmatrix} \boldsymbol{g}_{e_1}^{(1)} & \cdots & \boldsymbol{g}_{e_{N_e}}^{(1)} \\ \vdots & \ddots & \vdots \\ \boldsymbol{g}_{e_1}^{(N_c)} & \cdots & \boldsymbol{g}_{e_{N_e}}^{(N_c)} \end{bmatrix} = [\boldsymbol{g}_1, \ldots, \boldsymbol{g}_{N_e}], \qquad \boldsymbol{b} = \begin{bmatrix} \boldsymbol{b}^{(1)} \\ \vdots \\ \boldsymbol{b}^{(N_c)} \end{bmatrix}. \tag{25}$$

Here $\boldsymbol{G} \in \mathbb{R}^{nN_c \times N_e}$ and $\boldsymbol{b} \in \mathbb{R}^{nN_c}$. Column $\boldsymbol{g}_e$ contains the contributions of one entity across all configurations, and each configuration supplies n rows. In particular, $G_{[c-1]n+i,e} = g_{e,i}^{(c)}$. Since the target is the sum of all entity contributions, these quantities satisfy

$$\boldsymbol{G}\boldsymbol{1} = \boldsymbol{b}, \tag{26}$$

where $\boldsymbol{1} \in \mathbb{R}^{N_e}$ is the vector of ones.

The training task is therefore to replace the full sum by a sparse, nonnegative weight vector whose weighted contributions reproduce $\boldsymbol{b}$ to the required accuracy. This leads to the ideal sampling problem

$$\min_{\boldsymbol{\xi} \in \mathbb{R}^{N_e},\ \boldsymbol{\xi} \geq \boldsymbol{0}} \|\boldsymbol{\xi}\|_0 \quad \text{subject to} \quad \|\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}\|_2 \leq \tau \|\boldsymbol{b}\|_2, \qquad 0 < \tau < 1, \tag{27}$$

where $\|\boldsymbol{\xi}\|_0$ counts the nonzero weights and τ is the prescribed relative training tolerance. Each entry of $\boldsymbol{\xi}$ corresponds to one original mesh entity, and its positive support defines the reduced mesh:

$$\widetilde{\mathcal{E}} = \{e \in \mathcal{E} : \xi_e > 0\}, \qquad n_e = |\widetilde{\mathcal{E}}|. \tag{28}$$

Because the cardinality problem in Equation (27) is combinatorial, the sampled set is constructed progressively using nonnegative least-squares (NNLS) fits [6]. The fitting objective is

$$\min_{\boldsymbol{\xi} \geq \boldsymbol{0}} F(\boldsymbol{\xi}), \qquad F(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}\|_2^2. \tag{29}$$

The NNLS fits determine the weights on the selected entities, while progressive selection and stopping at the prescribed tolerance produce a sparse rule. Expanding the fitting objective gives

$$\begin{aligned} 2F(\boldsymbol{\xi}) &= [\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}]^T[\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}] \\ &= \boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{b} - \boldsymbol{b}^T\boldsymbol{G}\boldsymbol{\xi} + \boldsymbol{b}^T\boldsymbol{b} \\ &= \boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{G}\boldsymbol{\xi} - 2\boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{b} + \boldsymbol{b}^T\boldsymbol{b}. \end{aligned} \tag{30}$$

The two scalar cross terms are transposes of one another and hence are equal. Differentiation with respect to the weights yields

$$\nabla_{\boldsymbol{\xi}} F = \boldsymbol{G}^T \boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{G}^T \boldsymbol{b}. \tag{31}$$

This gradient provides the entity-selection criterion used to enlarge the sampled set.

At sampling iteration k, let $\mathcal{I}^k$ denote the working set of selected column indices and $\mathcal{Z}^k = \mathcal{E} \backslash \mathcal{I}^k$ the entities not yet selected. The weights indexed by $\mathcal{Z}^k$ remain fixed to zero, and only those indexed by $\mathcal{I}^k$ are varied in the NNLS refit. With these unselected weights held fixed, the current weight vector is obtained from

$$\boldsymbol{\xi}^k \in \underset{\boldsymbol{\xi} \geq \mathbf{0},\, \boldsymbol{\xi}_{\mathcal{Z}^k} = \mathbf{0}}{\arg\min} \frac{1}{2} \|\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}\|_2^2. \tag{32}$$

Writing $\boldsymbol{G}_{\mathcal{I}^k}$ for the selected columns, this refit can be computed on the smaller matrix as

$$\boldsymbol{\xi}^k_{\mathcal{I}^k} \in \underset{\boldsymbol{z} \geq \mathbf{0}}{\arg\min} \frac{1}{2} \|\boldsymbol{G}_{\mathcal{I}^k} \boldsymbol{z} - \boldsymbol{b}\|_2^2. \tag{33}$$

The coefficients of the unselected columns remain zero in the complete N_e-entry vector. Selected coefficients can increase, decrease, or become zero after a refit, so the working set can be larger than the positive support of the resulting weights.

To select the next entity, evaluate the cubature fitting residual and its correlations with the columns of $\boldsymbol{G}$:

$$\boldsymbol{h}^k = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^k, \qquad \boldsymbol{\mu}^k = \boldsymbol{G}^T \boldsymbol{h}^k, \qquad \mu_e^k = \boldsymbol{g}_e^T \boldsymbol{h}^k. \tag{34}$$

In this fitting problem, $\boldsymbol{h}^k$ measures the error in the cubature approximation, and $\boldsymbol{\mu}^k = -\nabla F(\boldsymbol{\xi}^k)$. To determine which entity to add, consider increasing one unselected weight while holding all other weights fixed. For $e \in \mathcal{Z}^k$, the current coefficient is $\xi_e^k = 0$, so a nonnegative increment is an admissible direction. Let $\boldsymbol{e}_e \in \mathbb{R}^{N_e}$ have entry one at entity e and zero elsewhere. The candidate vector $\boldsymbol{\xi}^k + t\boldsymbol{e}_e$, with $t \geq 0$, changes only this coefficient from zero to t.

Multiplication by $\boldsymbol{e}_e$ selects column e of the training matrix, so $\boldsymbol{G}\boldsymbol{e}_e = \boldsymbol{g}_e$. Using the fitting residual defined in Equation (34), the residual associated with the candidate weights becomes

$$\boldsymbol{b} - \boldsymbol{G}[\boldsymbol{\xi}^k + t\boldsymbol{e}_e] = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^k - t\boldsymbol{g}_e = \boldsymbol{h}^k - t\boldsymbol{g}_e.$$

Substitution into the objective and expansion of its squared norm give

$$\begin{aligned} F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e) &= \frac{1}{2} \|\boldsymbol{h}^k - t\boldsymbol{g}_e\|_2^2 \\ &= \frac{1}{2} [\boldsymbol{h}^k - t\boldsymbol{g}_e]^T [\boldsymbol{h}^k - t\boldsymbol{g}_e] \\ &= \frac{1}{2} \|\boldsymbol{h}^k\|_2^2 - \frac{t}{2} \left[[\boldsymbol{h}^k]^T \boldsymbol{g}_e + \boldsymbol{g}_e^T \boldsymbol{h}^k \right] + \frac{t^2}{2} \|\boldsymbol{g}_e\|_2^2 \\ &= F(\boldsymbol{\xi}^k) - t\mu_e^k + \frac{t^2}{2} \|\boldsymbol{g}_e\|_2^2. \end{aligned} \tag{35}$$

Here the two mixed products are equal scalars, $\frac{1}{2}\|\boldsymbol{h}^k\|_2^2 = F(\boldsymbol{\xi}^k)$, and $\boldsymbol{g}_e^T \boldsymbol{h}^k = \mu_e^k$. These identities give the last line of Equation (35).

During this candidate comparison, $\boldsymbol{\xi}^k$, $\boldsymbol{h}^k$, $\boldsymbol{g}_e$, and μ_e^k are fixed; only t varies. Differentiating the constant, linear, and quadratic terms therefore gives

$$\frac{d}{dt}F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e) = -\mu_e^k + t\|\boldsymbol{g}_e\|_2^2.$$

At $t = 0$, the second term vanishes. The chain rule expresses the same derivative as the gradient of F multiplied by the weight direction $\boldsymbol{e}_e$, yielding

$$\frac{d}{dt}F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e)\bigg|_{t=0} = \nabla F(\boldsymbol{\xi}^k)^T\boldsymbol{e}_e = -\mu_e^k. \tag{36}$$

Thus a positive correlation gives a negative directional derivative: increasing that entity's weight initially decreases the fitting error. Comparing candidates using the same small weight increment, a larger μ_e^k gives a greater initial decrease. Selecting the most negative directional derivative therefore yields the greedy rule

$$\begin{aligned} e^k &\in \arg\min_{e\in\mathcal{Z}^k} \frac{d}{dt}F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e)\bigg|_{t=0} \\ &= \arg\max_{e\in\mathcal{Z}^k} \mu_e^k. \end{aligned} \tag{37}$$

This comparison concerns the slope at $t = 0$ and does not guarantee the largest decrease after refitting. Once e^k is added to the working set, all selected weights are refitted against the original target $\boldsymbol{b}$ using Equation (33). The NNLS refit determines their final values for this sampling iteration; the candidate increment above is used to choose the entity. This selection and refitting process continues until

$$\eta = \frac{\|\boldsymbol{G}\boldsymbol{\xi}^k - \boldsymbol{b}\|_2}{\|\boldsymbol{b}\|_2} \le \tau. \tag{38}$$

Algorithm 1 summarizes the procedure, starting from an empty working set and zero weights. Its NNLS refits can be computed using active-set methods and factorization updates [6, Sections 4–5]. The last iterate defines the trained weights $\boldsymbol{\xi}$, and the reduced mesh is their positive support as in Equation (28).

Algorithm 1 Classical ECSW mesh sampling by NNLS

Input: $\boldsymbol{G} = [\boldsymbol{g}_e]$, $\boldsymbol{b} \neq \boldsymbol{0}$, tolerance $\tau \in (0, 1)$
Output: Weights $\boldsymbol{\xi} \geq \boldsymbol{0}$ and reduced mesh $\widetilde{\mathcal{E}}$

$k \leftarrow 0$
$\mathcal{I}^0 \leftarrow \varnothing$; $\mathcal{Z}^0 \leftarrow \mathcal{E}$
$\boldsymbol{\xi}^0 \leftarrow \boldsymbol{0}$
$\boldsymbol{h}^0 \leftarrow \boldsymbol{b}$
while $\|\boldsymbol{h}^k\|_2 > \tau \|\boldsymbol{b}\|_2$ **do**
    Compute the correlations: $\boldsymbol{\mu}^k \leftarrow \boldsymbol{G}^T \boldsymbol{h}^k$
    Select the most correlated entity: $e^k \leftarrow \arg\max_{e \in \mathcal{Z}^k} \mu_e^k$
    Add the candidate: $\mathcal{I}^{k+1} \leftarrow \mathcal{I}^k \cup \{e^k\}$
    Update the unselected set: $\mathcal{Z}^{k+1} \leftarrow \mathcal{E} \setminus \mathcal{I}^{k+1}$
    Refit the weights of all selected entities by NNLS:
    $\boldsymbol{\xi}_{\mathcal{I}^{k+1}}^{k+1} \in \arg\min_{\boldsymbol{z} \geq \boldsymbol{0}} \frac{1}{2} \|\boldsymbol{G}_{\mathcal{I}^{k+1}} \boldsymbol{z} - \boldsymbol{b}\|_2^2$
    Set unselected weights to zero: $\boldsymbol{\xi}_{\mathcal{Z}^{k+1}}^{k+1} \leftarrow \boldsymbol{0}$
    Recompute the full fitting residual: $\boldsymbol{h}^{k+1} \leftarrow \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^{k+1}$
    $k \leftarrow k + 1$
end while
$\widetilde{\mathcal{E}} \leftarrow \{e \in \mathcal{E} : \xi_e^k > 0\}$
return $\boldsymbol{\xi}^k$ and $\widetilde{\mathcal{E}}$

2.2 Online solution

With the sampled mesh and its weights fixed, the online problem is to solve $\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}^{m+1}) = \boldsymbol{0}$ at each time step. For LSPG, the relation $\boldsymbol{W}_e = \partial \boldsymbol{r}_e / \partial \boldsymbol{q}$ in Equation (17) associates these reduced equations with the sampled least-squares objective

$$\Phi_{\mathrm{ECSW}}(\boldsymbol{q}) = \frac{1}{2} \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2. \tag{39}$$

Indeed, differentiation with the cubature weights held fixed gives

$$\nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e(\boldsymbol{q})^T \boldsymbol{r}_e(\boldsymbol{q}) = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}). \tag{40}$$

The classical ECSW equations are therefore the stationarity conditions of a single weighted least-squares problem. This interpretation follows from using the same entity weight in every reduced equation.

The coordinates can consequently be computed using the Gauss–Newton construction of Section 1.1. At the current iterate, each selected residual block is linearized as $\boldsymbol{r}_e(\boldsymbol{q} + \Delta\boldsymbol{q}) \approx \boldsymbol{r}_e(\boldsymbol{q}) + \boldsymbol{W}_e(\boldsymbol{q})\Delta\boldsymbol{q}$. Substituting these linearizations into Equation (39) gives the least-squares problem for the coordinate increment

$$\min_{\Delta\boldsymbol{q} \in \mathbb{R}^n} \frac{1}{2} \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e + \boldsymbol{W}_e \Delta\boldsymbol{q}\|_2^2, \tag{41}$$

whose normal equations are

$$\left[\sum_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{W}_e \right] \Delta\boldsymbol{q} = - \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e. \tag{42}$$

Both sides are assembled from the selected residual and test-basis blocks evaluated at the current reduced state. Solving this $n \times n$ system gives the coordinate correction; after the update, the state is reconstructed on the augmented mesh using Equation (21), and the local blocks are evaluated again for the next iteration.

At a fixed time level, let k denote the nonlinear iteration and $\boldsymbol{q}^{(0)}$ the initial guess. The ROB, reference state, sampled entities, cubature weights, and known history remain fixed throughout this solve. A relative convergence criterion for the reduced stationarity equations is

$$\|\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}^{(k)})\|_2 \le \varepsilon_{\mathrm{NL}} \|\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}^{(0)})\|_2, \qquad 0 < \varepsilon_{\mathrm{NL}} < 1. \tag{43}$$

This criterion monitors the gradient of the sampled objective, whose local HDM residuals can remain nonzero at a least-squares solution. Algorithm 2 summarizes the resulting full-step Gauss–Newton iteration.

Algorithm 2 Online solution of HPROM (ECSW)

Input: Current time level t^{m+1}, prescribed parameters $\boldsymbol{\mu}$, known approximate history states on $\widetilde{\mathcal{E}}^{+}$, restricted basis $\boldsymbol{V}^{+}$, reference state $\boldsymbol{u}_{\mathrm{ref}}^{+}$, selected mesh $\widetilde{\mathcal{E}}$, weights $\{\xi_e\}_{e\in\widetilde{\mathcal{E}}}$, initial coordinates $\boldsymbol{q}^{(0)}$, and tolerance $\varepsilon_{\mathrm{NL}} \in (0,1)$

Output: Solution coordinates $\boldsymbol{q}^{m+1}$ at the current time step

$k \leftarrow 0$
loop
  $\widetilde{\boldsymbol{u}}^{+,(k)} \leftarrow \boldsymbol{u}_{\mathrm{ref}}^{+} + \boldsymbol{V}^{+}\boldsymbol{q}^{(k)}$
  for all $e \in \widetilde{\mathcal{E}}$ **do**
    Evaluate $\boldsymbol{r}_e^{(k)}$ at the reconstructed state $\widetilde{\boldsymbol{u}}^{+,(k)}$
    $\boldsymbol{W}_e^{(k)} \leftarrow \dfrac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}(\boldsymbol{q}^{(k)})$
  end for
  $\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^{(k)}]^T \boldsymbol{r}_e^{(k)}$
  if $k = 0$ **then**
    $\rho_0 \leftarrow \|\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},(0)}\|_2$
  end if
  if $\|\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},(k)}\|_2 \le \varepsilon_{\mathrm{NL}}\rho_0$ **then**
    return $\boldsymbol{q}^{m+1} \leftarrow \boldsymbol{q}^{(k)}$
  end if
  $\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^{(k)}]^T \boldsymbol{W}_e^{(k)}$
  Solve the reduced linear system for $\Delta\boldsymbol{q}^{(k)}$:
  $\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW},(k)} \Delta\boldsymbol{q}^{(k)} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},(k)}$
  $\boldsymbol{q}^{(k+1)} \leftarrow \boldsymbol{q}^{(k)} + \Delta\boldsymbol{q}^{(k)}$
  $k \leftarrow k + 1$
end loop

3 By-mode energy-conserving sampling and weighting (BM-ECSW)

In the classical ECSW approximation, each selected entity contributes to all reduced equations through a single cubature weight. A single weight vector must therefore fit the projected residual across all training configurations and reduced components. By-mode ECSW (BM-ECSW) [9] extends this construction by assigning a separate weight to each reduced component while retaining one shared sampled mesh. This additional freedom in the offline fit is intended to reduce the number of entities needed to attain a prescribed training accuracy.

To express this extension, write the classical approximation in Equation (18) componentwise as

$$\widetilde{r}_{n,i}^{\mathrm{ECSW}}(\boldsymbol{q}) = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e g_{e,i}(\boldsymbol{q}), \qquad i = 1,\dots,n. \tag{44}$$

For each entity e, replace the scalar ξ_e by nonnegative coefficients $\xi_{e,i}$ associated with the reduced equations, and collect them in the diagonal matrix

$$\boldsymbol{D}_e = \mathrm{diag}(\xi_{e,1},\dots,\xi_{e,n}). \tag{45}$$

The resulting cubature rule approximates the full projected residual by

$$\begin{aligned} \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}) &= \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{g}_e(\boldsymbol{q}), \\ \widetilde{r}_{n,i}^{\mathrm{BM}}(\boldsymbol{q}) &= \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} g_{e,i}(\boldsymbol{q}), \qquad i = 1,\dots,n. \end{aligned} \tag{46}$$

Here $\boldsymbol{g}_e = \boldsymbol{W}_e^T \boldsymbol{r}_e$ is the projected entity contribution defined in Equation (17). Thus BM-ECSW evaluates the same local residual and test-basis blocks as classical ECSW, but weights their reduced components individually. An entity belongs to the shared sampled mesh if at least one of its modal weights is positive; its contribution to other reduced equations can have zero weight. The state entries required to evaluate these selected contributions define the augmented mesh through the stencil operators introduced in Section 2.

Once trained, the modal weights remain fixed throughout the online simulation. Taking $\xi_{e,i} = \xi_e$ for all i gives $\boldsymbol{D}_e = \xi_e \boldsymbol{I}_n$ and recovers the classical rule. Consequently, on a fixed sampled set, the admissible BM weights include every classical weight vector, and the minimum aggregate training error cannot increase when the weights are allowed to differ between modes.

3.1 Offline mesh sampling and modal weight fitting

The BM cubature rule is trained using the same projected contributions and configurations as the classical rule. In the matrix $\boldsymbol{G}$ of Equation (25), each configuration supplies one row per reduced equation. Collecting the rows associated with equation i over all N_c configurations gives the modal training matrix and its target:

$$\boldsymbol{G}^{[i]} = \begin{bmatrix} g_{e_1,i}^{(1)} & \cdots & g_{e_{N_e},i}^{(1)} \\ \vdots & \ddots & \vdots \\ g_{e_1,i}^{(N_c)} & \cdots & g_{e_{N_e},i}^{(N_c)} \end{bmatrix}, \qquad \boldsymbol{b}^{[i]} = \boldsymbol{G}^{[i]}\mathbf{1}. \tag{47}$$

Here $\boldsymbol{G}^{[i]} \in \mathbb{R}^{N_c \times N_e}$, $\boldsymbol{b}^{[i]} \in \mathbb{R}^{N_c}$, and column $\boldsymbol{g}_e^{[i]}$ contains the contributions of entity e to equation i at all training configurations. These matrices partition the rows of $\boldsymbol{G}$ by reduced component, so both cubature rules use the same projected training data.

Each mode is assigned a weight vector $\boldsymbol{\xi}^{[i]} \in \mathbb{R}^{N_e}$. Collecting these vectors yields

$$\boldsymbol{\Xi} = [\boldsymbol{\xi}^{[1]}, \ldots, \boldsymbol{\xi}^{[n]}] \in \mathbb{R}^{N_e \times n}, \qquad \Xi_{e,i} = \xi_{e,i}, \tag{48}$$

whose rows correspond to entities and whose columns correspond to modes. Evaluating an entity is necessary whenever at least one of its modal weights is positive. The BM counterpart of the ideal classical sampling problem in Equation (27) seeks nonnegative modal weights that meet the prescribed accuracy using as few sampled entities as possible:

$$\min_{\boldsymbol{\Xi} \geq 0} \|\boldsymbol{\Xi}\|_{\text{row}-0} \quad \text{subject to} \quad \sum_{i=1}^{n} \|\boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i]} - \boldsymbol{b}^{[i]}\|_2^2 \leq \tau^2 \sum_{i=1}^{n} \|\boldsymbol{b}^{[i]}\|_2^2, \tag{49}$$

where τ is the relative training tolerance. The row-ℓ_0 quasi-norm $\|\boldsymbol{\Xi}\|_{\text{row}-0}$ counts the rows containing at least one nonzero weight and therefore equals the number of sampled entities, $|\widetilde{\mathcal{E}}|$ [10]. The modal weights of a selected entity can differ and can be zero in some modes.

As in classical ECSW, a sparse rule is constructed progressively by selecting entities and refitting their weights using NNLS. The aggregate fitting objective is

$$F_{\text{BM}}(\boldsymbol{\Xi}) = \frac{1}{2} \sum_{i=1}^{n} \|\boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i]} - \boldsymbol{b}^{[i]}\|_2^2. \tag{50}$$

Expanding each modal term gives

$$\begin{aligned} F_{\text{BM}}(\boldsymbol{\Xi}) = \frac{1}{2} \sum_{i} \Big[& [\boldsymbol{\xi}^{[i]}]^T [\boldsymbol{G}^{[i]}]^T \boldsymbol{G}^{[i]} \boldsymbol{\xi}^{[i]} \\ & - 2[\boldsymbol{\xi}^{[i]}]^T [\boldsymbol{G}^{[i]}]^T \boldsymbol{b}^{[i]} + \|\boldsymbol{b}^{[i]}\|_2^2 \Big]. \end{aligned} \tag{51}$$

Because the modal matrices collect disjoint rows of the classical training matrix, their quadratic and linear terms satisfy

$$\begin{aligned} \boldsymbol{G}^T\boldsymbol{G} &= \sum_{i=1}^{n} [\boldsymbol{G}^{[i]}]^T \boldsymbol{G}^{[i]}, \\ \boldsymbol{G}^T\boldsymbol{b} &= \sum_{i=1}^{n} [\boldsymbol{G}^{[i]}]^T \boldsymbol{b}^{[i]}, \qquad \|\boldsymbol{b}\|_2^2 = \sum_{i=1}^{n} \|\boldsymbol{b}^{[i]}\|_2^2. \end{aligned} \tag{52}$$

Substituting a common weight vector $\boldsymbol{\xi}^{[i]} = \boldsymbol{\xi}$ into Equation (51) therefore recovers the classical fitting objective in Equation (30). The BM fit retains the same data and total error measure while allowing a different weight vector in each modal term.

At sampling iteration k, all modes use the same working set $\mathcal{I}^k$, and $\mathcal{Z}^k = \mathcal{E} \setminus \mathcal{I}^k$ contains the entities not yet selected. The weights indexed by $\mathcal{Z}^k$ remain fixed to zero. With these entries held fixed, the selected weights are obtained from the modal NNLS problems

$$\boldsymbol{\xi}^{[i],k} \in \underset{\boldsymbol{\xi} \geq \mathbf{0},\, \boldsymbol{\xi}_{\mathcal{Z}^k} = \mathbf{0}}{\arg\min} \frac{1}{2} \|\boldsymbol{G}^{[i]}\boldsymbol{\xi} - \boldsymbol{b}^{[i]}\|_2^2, \qquad i = 1, \ldots, n. \tag{53}$$

Each fitted vector has N_e entries, but its selected coefficients can be computed using the smaller matrix $\boldsymbol{G}^{[i]}_{\mathcal{I}^k}$. Once the shared working set is prescribed, these n fits are independent. Their cubature fitting residuals and column correlations are

$$\boldsymbol{h}^{[i],k} = \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k}, \qquad \boldsymbol{\mu}^{[i],k} = [\boldsymbol{G}^{[i]}]^T\boldsymbol{h}^{[i],k}. \tag{54}$$

Differentiating the aggregate objective with respect to the weights of mode i involves only its modal term, since the other terms do not depend on $\boldsymbol{\xi}^{[i]}$. At the current iterate, this gives

$$\nabla_{\boldsymbol{\xi}^{[i]}} F_{\mathrm{BM}}\Big|_{\boldsymbol{\Xi}^k} = [\boldsymbol{G}^{[i]}]^T[\boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k} - \boldsymbol{b}^{[i]}] = -\boldsymbol{\mu}^{[i],k}.$$

Thus each modal correlation vector is the negative gradient with respect to that mode's weights. To select one entity for the shared mesh, these correlations are combined into

$$\boldsymbol{\mu}^{\,k} = \sum_{i=1}^{n} \boldsymbol{\mu}^{[i],k}, \qquad \mu_e^k = \sum_{i=1}^{n} [\boldsymbol{g}_e^{[i]}]^T\boldsymbol{h}^{[i],k}. \tag{55}$$

To derive the selection criterion, consider an entity $e \in \mathcal{Z}^k$, whose weights are currently zero in every mode. Increase all of its modal weights by the same $t \geq 0$, holding every other row of the weight matrix fixed:

$$\boldsymbol{\Xi}(t) = \boldsymbol{\Xi}^k + t\boldsymbol{e}_e\mathbf{1}_n^T. \tag{56}$$

Here $\boldsymbol{e}_e \in \mathbb{R}^{N_e}$ has entry one at entity e and zero elsewhere, and $\mathbf{1}_n \in \mathbb{R}^n$ is the vector of ones. The matrix $\boldsymbol{e}_e\mathbf{1}_n^T$ has ones along row e and zeros elsewhere, so the weight vector in each column becomes

$$\boldsymbol{\xi}^{[i],k} + t\boldsymbol{e}_e, \qquad i = 1, \ldots, n.$$

Multiplication by $\boldsymbol{e}_e$ selects column e of each modal training matrix, giving $\boldsymbol{G}^{[i]}\boldsymbol{e}_e = \boldsymbol{g}_e^{[i]}$. The fitting residual for mode i therefore changes according to

$$\begin{aligned} \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}[\boldsymbol{\xi}^{[i],k} + t\boldsymbol{e}_e] &= \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k} - t\boldsymbol{g}_e^{[i]} \\ &= \boldsymbol{h}^{[i],k} - t\boldsymbol{g}_e^{[i]}. \end{aligned}$$

Substituting these residuals into the aggregate objective and expanding the squared norms yield

$$\begin{aligned} F_{\mathrm{BM}}(\boldsymbol{\Xi}(t)) &= \frac{1}{2}\sum_i \|\boldsymbol{h}^{[i],k} - t\boldsymbol{g}_e^{[i]}\|_2^2 \\ &= \frac{1}{2}\sum_i [\boldsymbol{h}^{[i],k} - t\boldsymbol{g}_e^{[i]}]^T[\boldsymbol{h}^{[i],k} - t\boldsymbol{g}_e^{[i]}] \\ &= \frac{1}{2}\sum_i \|\boldsymbol{h}^{[i],k}\|_2^2 - \frac{t}{2}\sum_i \left[[\boldsymbol{h}^{[i],k}]^T\boldsymbol{g}_e^{[i]} + [\boldsymbol{g}_e^{[i]}]^T\boldsymbol{h}^{[i],k}\right] \\ &\quad + \frac{t^2}{2}\sum_i \|\boldsymbol{g}_e^{[i]}\|_2^2 \\ &= F_{\mathrm{BM}}(\boldsymbol{\Xi}^k) - t\mu_e^k + \frac{t^2}{2}\sum_i \|\boldsymbol{g}_e^{[i]}\|_2^2. \end{aligned} \tag{57}$$

For each mode, the two mixed products are equal scalars. Combining each pair gives twice its modal correlation, so summing these terms gives $-t\mu_e^k$ by Equation (55). The constant term is the current fitting error $F_{\mathrm{BM}}(\boldsymbol{\Xi}^k)$. These identities give the last line of Equation (57). This

expansion has the same form as its classical counterpart in Equation (35), with $\sum_{i=1}^{n} \|\boldsymbol{g}_e^{[i]}\|_2^2 = \|\boldsymbol{g}_e\|_2^2$.

Differentiating Equation (57) with respect to t gives

$$\frac{d}{dt} F_{\mathrm{BM}}(\boldsymbol{\Xi}(t)) = -\mu_e^k + t \sum_i \|\boldsymbol{g}_e^{[i]}\|_2^2.$$

At $t = 0$, the second term vanishes, leaving

$$\left.\frac{d}{dt} F_{\mathrm{BM}}(\boldsymbol{\Xi}(t))\right|_{t=0} = -\mu_e^k. \tag{58}$$

A positive score therefore gives an initial decrease along the candidate direction in Equation (56). Selecting the most negative slope yields

$$\begin{aligned} e^k &\in \arg\min_{e \in \mathcal{Z}^k} \left.\frac{d}{dt} F_{\mathrm{BM}}(\boldsymbol{\Xi}^k + t\boldsymbol{e}_e \mathbf{1}_n^T)\right|_{t=0} \\ &= \arg\max_{e \in \mathcal{Z}^k} \mu_e^k. \end{aligned} \tag{59}$$

The signed modal correlations can cancel in their sum. As in classical ECSW, ranking candidates by their initial slopes does not guarantee the largest reduction after refitting.

When $\boldsymbol{\xi}^{[i],k} = \boldsymbol{\xi}^k$ for all i, the modal residuals are the corresponding row blocks of the classical fitting residual. The score then recovers the classical correlation in Equation (34):

$$\mu_e^k = \sum_{i=1}^{n} [\boldsymbol{g}_e^{[i]}]^T \boldsymbol{h}^{[i],k} = \boldsymbol{g}_e^T \boldsymbol{h}^k.$$

This condition holds at zero initialization, so both sampling procedures begin with identical candidate scores.

After each candidate is added, all selected weights are refitted independently for each mode against the original targets $\boldsymbol{b}^{[i]}$. The resulting modal residuals can lead to different entity selections from the classical procedure in subsequent iterations. Selection and refitting continue until the aggregate relative error satisfies

$$\eta^2 = \frac{\sum_i \|\boldsymbol{h}^{[i],k}\|_2^2}{\sum_i \|\boldsymbol{b}^{[i]}\|_2^2} \leq \tau^2. \tag{60}$$

Algorithm 3 summarizes this procedure. The final modal iterates define the trained weights $\boldsymbol{\xi}^{[i]}$, and their combined positive support defines the shared sampled mesh. These weights and the corresponding augmented mesh supply the data needed for the online reduced solve.

Algorithm 3 BM-ECSW mesh sampling by modal NNLS

Input: $\boldsymbol{G}^{[i]} = [\boldsymbol{g}_e^{[i]}]$, $\boldsymbol{b}^{[i]}$ with $\sum_i \|\boldsymbol{b}^{[i]}\|_2^2 > 0$, tolerance $\tau \in (0,1)$
Output: Modal weights $\boldsymbol{\xi}^{[i]} \geq \boldsymbol{0}$ and reduced mesh $\widetilde{\mathcal{E}}$

$k \leftarrow 0$; $\mathcal{I}^0 \leftarrow \varnothing$; $\mathcal{Z}^0 \leftarrow \mathcal{E}$
$\boldsymbol{\xi}^{[i],0} \leftarrow \boldsymbol{0}$; $\boldsymbol{h}^{[i],0} \leftarrow \boldsymbol{b}^{[i]}$ for every i
while $\sum_i \|\boldsymbol{h}^{[i],k}\|_2^2 > \tau^2 \sum_i \|\boldsymbol{b}^{[i]}\|_2^2$ **do**
    Sum the modal correlations: $\boldsymbol{\mu}^k \leftarrow \sum_{i=1}^{n} [\boldsymbol{G}^{[i]}]^T \boldsymbol{h}^{[i],k}$
    Select the most correlated entity: $e^k \leftarrow \arg\max_{e \in \mathcal{Z}^k} \mu_e^k$
    Add the candidate: $\mathcal{I}^{k+1} \leftarrow \mathcal{I}^k \cup \{e^k\}$
    Update the unselected set: $\mathcal{Z}^{k+1} \leftarrow \mathcal{E} \setminus \mathcal{I}^{k+1}$
    for $i = 1, \ldots, n$ **do**
        Refit the selected weights of mode i by NNLS:
        $\boldsymbol{\xi}_{\mathcal{I}^{k+1}}^{[i],k+1} \in \arg\min_{\boldsymbol{z} \geq \boldsymbol{0}} \frac{1}{2} \|\boldsymbol{G}_{\mathcal{I}^{k+1}}^{[i]} \boldsymbol{z} - \boldsymbol{b}^{[i]}\|_2^2$
        Set unselected weights to zero: $\boldsymbol{\xi}_{\mathcal{Z}^{k+1}}^{[i],k+1} \leftarrow \boldsymbol{0}$
        Recompute the fitting residual: $\boldsymbol{h}^{[i],k+1} \leftarrow \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]} \boldsymbol{\xi}^{[i],k+1}$
    end for
    $k \leftarrow k + 1$
end while
$\widetilde{\mathcal{E}} \leftarrow \bigcup_{i=1}^{n} \{e \in \mathcal{E} : \xi_{e,i}^k > 0\}$
return $\{\boldsymbol{\xi}^{[i],k}\}_{i=1}^n$ and $\widetilde{\mathcal{E}}$

3.2 Online solution

With the trained mesh and modal weights fixed, the generalized coordinates at each time step are determined by solving

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}^{m+1}) = \boldsymbol{0}. \tag{61}$$

These equations approximate the full LSPG projected equations using the cubature rule in Equation (46). Their relation to least-squares minimization follows by associating the weights of each reduced equation i with the function

$$\Phi_i(\boldsymbol{q}) = \frac{1}{2} \sum_{e \in \widetilde{\mathcal{E}}} \xi_{e,i} \|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2, \qquad \widetilde{r}_{n,i}^{\mathrm{BM}} = \frac{\partial \Phi_i}{\partial q_i}. \tag{62}$$

This identity follows from $\boldsymbol{W}_e = \partial \boldsymbol{r}_e / \partial \boldsymbol{q}$ with the weights held fixed. In classical ECSW, all Φ_i coincide with the sampled objective in Equation (39). For independently trained modal weights, these functions can differ, and the indicated partial derivatives are not guaranteed to form the gradient of one common weighted least-squares objective. Appendix A develops this distinction and its implications for symmetry and conservative structural dynamics. The online iteration is therefore constructed directly from the weighted projected equations.

At the current iterate $\boldsymbol{q}$, evaluate the selected local residuals and test-basis blocks using the state restricted to the augmented mesh. As in the classical Gauss–Newton construction, approximate each local residual in terms of a coordinate increment $\Delta \boldsymbol{q}$ by

$$\widehat{\boldsymbol{r}}_e(\Delta \boldsymbol{q}) = \boldsymbol{r}_e + \boldsymbol{W}_e \Delta \boldsymbol{q}, \qquad \boldsymbol{r}_e(\boldsymbol{q} + \Delta \boldsymbol{q}) \approx \widehat{\boldsymbol{r}}_e(\Delta \boldsymbol{q}). \tag{63}$$

The test-basis blocks are evaluated at $\boldsymbol{q}$ and held fixed while computing the increment. Applying the weights of reduced equation i to the projected linearized contributions gives

$$\begin{aligned}\widehat{r}_{n,i}^{\mathrm{BM}}(\Delta\boldsymbol{q}) &= \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i}\boldsymbol{W}_e(:,i)^T\left[\boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}\right] \\ &= \widetilde{r}_{n,i}^{\mathrm{BM}}(\boldsymbol{q}) + \sum_{j=1}^{n}\left[\sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i}\boldsymbol{W}_e(:,i)^T\boldsymbol{W}_e(:,j)\right]\Delta q_j.\end{aligned} \tag{64}$$

Here $\boldsymbol{W}_e(:,i)$ is the derivative of the local residual with respect to q_i. Collecting the coefficients of all coordinate increments in Equation (64) yields the reduced update matrix

$$\begin{aligned}\widehat{\boldsymbol{J}}_n^{\mathrm{BM}} &= \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e\boldsymbol{W}_e^T\boldsymbol{W}_e \in \mathbb{R}^{n\times n}, \\ \left[\widehat{J}_n^{\mathrm{BM}}\right]_{ij} &= \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i}\sum_{\ell=1}^{d_e} W_{e,\ell,i}W_{e,\ell,j}.\end{aligned} \tag{65}$$

The index ℓ runs over the local HDM residual entries, while i and j identify the reduced equation and coordinate. Setting the projected linearized equations to zero then gives the Gauss–Newton-type correction at iteration k of time step $m+1$:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}(\boldsymbol{q}^{m+1,k})\Delta\boldsymbol{q}^{m+1,k} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}^{m+1,k}). \tag{66}$$

Both sides reuse the selected local residual and test-basis blocks evaluated at the current reduced state. This $n\times n$ system computes all coordinate increments together, since each reduced equation generally depends on every coordinate. Holding the test-basis blocks fixed omits the residual-weighted second-derivative terms of the exact BM Jacobian, so the correction requires only local residuals and their first derivatives.

The placement of $\boldsymbol{D}_e$ in Equation (65) identifies the change from classical ECSW: row i uses $\xi_{e,i}$, whereas row j uses $\xi_{e,j}$. The entries (i,j) and (j,i) can differ, even though each local product $\boldsymbol{W}_e^T\boldsymbol{W}_e$ is symmetric. The BM update matrix can therefore be nonsymmetric. When $\xi_{e,i} = \xi_e$ for every mode, $\boldsymbol{D}_e = \xi_e\boldsymbol{I}_n$, all Φ_i reduce to the common sampled objective, and Equation (66) recovers the classical Gauss–Newton system in Equation (42). This restriction still permits different weights between entities.

After solving for the correction, the generalized coordinates are updated, the state is reconstructed on the augmented mesh using Equation (21), and the selected residual and test-basis blocks are evaluated again. At a fixed time level, let $\boldsymbol{q}^{(0)}$ denote the initial guess and k the nonlinear iteration index. The iteration continues until the BM projected equations satisfy the relative criterion

$$\|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}^{(k)})\|_2 \le \varepsilon_{\mathrm{NL}}\|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}^{(0)})\|_2. \tag{67}$$

This test has the same form as the classical criterion in Equation (43), with the projected residual replaced by its BM approximation. Algorithm 4 summarizes the resulting full-step iteration.

Algorithm 4 Online solution of HPROM (BM-ECSW)

Input: Current time level t^{m+1}, prescribed parameters $\boldsymbol{\mu}$, known approximate history states on $\widetilde{\mathcal{E}}^{+}$, restricted basis $\boldsymbol{V}^{+}$, reference state $\boldsymbol{u}^{+}_{\mathrm{ref}}$, selected mesh $\widetilde{\mathcal{E}}$, modal weight matrices $\{\boldsymbol{D}_e\}_{e\in\widetilde{\mathcal{E}}}$, initial coordinates $\boldsymbol{q}^{(0)}$, and tolerance $\varepsilon_{\mathrm{NL}} \in (0,1)$

Output: Solution coordinates $\boldsymbol{q}^{m+1}$ at the current time step

$k \leftarrow 0$
loop
  $\widetilde{\boldsymbol{u}}^{+,(k)} \leftarrow \boldsymbol{u}^{+}_{\mathrm{ref}} + \boldsymbol{V}^{+}\boldsymbol{q}^{(k)}$
  for all $e \in \widetilde{\mathcal{E}}$ **do**
    Evaluate $\boldsymbol{r}_e^{(k)}$ at the reconstructed state $\widetilde{\boldsymbol{u}}^{+,(k)}$
    $\boldsymbol{W}_e^{(k)} \leftarrow \dfrac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}(\boldsymbol{q}^{(k)})$
  end for
  $\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e[\boldsymbol{W}_e^{(k)}]^T \boldsymbol{r}_e^{(k)}$
  if $k = 0$ **then**
    $\rho_0 \leftarrow \|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(0)}\|_2$
  end if
  if $\|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)}\|_2 \le \varepsilon_{\mathrm{NL}}\rho_0$ **then**
    return $\boldsymbol{q}^{m+1} \leftarrow \boldsymbol{q}^{(k)}$
  end if
  $\widehat{\boldsymbol{J}}_n^{\mathrm{BM},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e[\boldsymbol{W}_e^{(k)}]^T \boldsymbol{W}_e^{(k)}$
  Solve the reduced linear system for $\Delta\boldsymbol{q}^{(k)}$:
  $\widehat{\boldsymbol{J}}_n^{\mathrm{BM},(k)} \Delta\boldsymbol{q}^{(k)} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)}$
  $\boldsymbol{q}^{(k+1)} \leftarrow \boldsymbol{q}^{(k)} + \Delta\boldsymbol{q}^{(k)}$
  $k \leftarrow k+1$
end loop

4 Two-dimensional parametric inviscid Burgers problem

The accuracy and computational efficiency of classical ECSW and BM-ECSW are assessed using the two-dimensional parametric inviscid Burgers problem [8, 1]. Its solution contains moving discontinuities that challenge affine reduced approximations. The two hyperreduction methods are applied to a common affine trial space and trained using the same projected state pairs, allowing their resulting mesh sizes, training costs, and online performance to be compared.

The governing equations are written in flux form as

$$\frac{\partial \boldsymbol{u}}{\partial t} + \frac{\partial \boldsymbol{F}(\boldsymbol{u})}{\partial x} + \frac{\partial \boldsymbol{G}(\boldsymbol{u})}{\partial y} = \boldsymbol{S}(x;\boldsymbol{\mu}), \qquad \boldsymbol{u} = \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \tag{68}$$

where u_x and u_y are the velocity components, and the flux vectors and source term are

$$\boldsymbol{F}(\boldsymbol{u}) = \begin{pmatrix} u_x^2/2 \\ u_x u_y/2 \end{pmatrix}, \qquad \boldsymbol{G}(\boldsymbol{u}) = \begin{pmatrix} u_x u_y/2 \\ u_y^2/2 \end{pmatrix}, \qquad \boldsymbol{S}(x;\boldsymbol{\mu}) = \begin{pmatrix} 0.02\exp(\mu_2 x) \\ 0 \end{pmatrix}. \tag{69}$$

The problem is posed on the square domain $\Omega = [0,100]^2$, with parameters $\boldsymbol{\mu} = (\mu_1, \mu_2)$ in $\mathcal{D} = [4.25, 5.50] \times [0.015, 0.030]$. The parameter μ_1 specifies the left inflow state $(\mu_1, 0)^T$, while

μ_2 controls the spatial variation of the source term. The initial velocities are $u_x = u_y = 1$; the bottom boundary has zero incoming flux, and outflow conditions are used at the top and right boundaries.

For spatial discretization, the domain is partitioned into a uniform 250×250 mesh and a cell-centered finite-volume method with first-order upwind fluxes is applied. The resulting HDM has $N_e = 62\,500$ cells and $N = 125\,000$ state entries, with two velocity unknowns per cell. Its semi-discrete equations are integrated using the implicit trapezoidal rule with $\Delta t = 0.05$, giving 500 steps over the interval $[0, T_f]$, where $T_f = 25$. Figure 1 illustrates the HDM solution and its centerline profiles at an intermediate time.

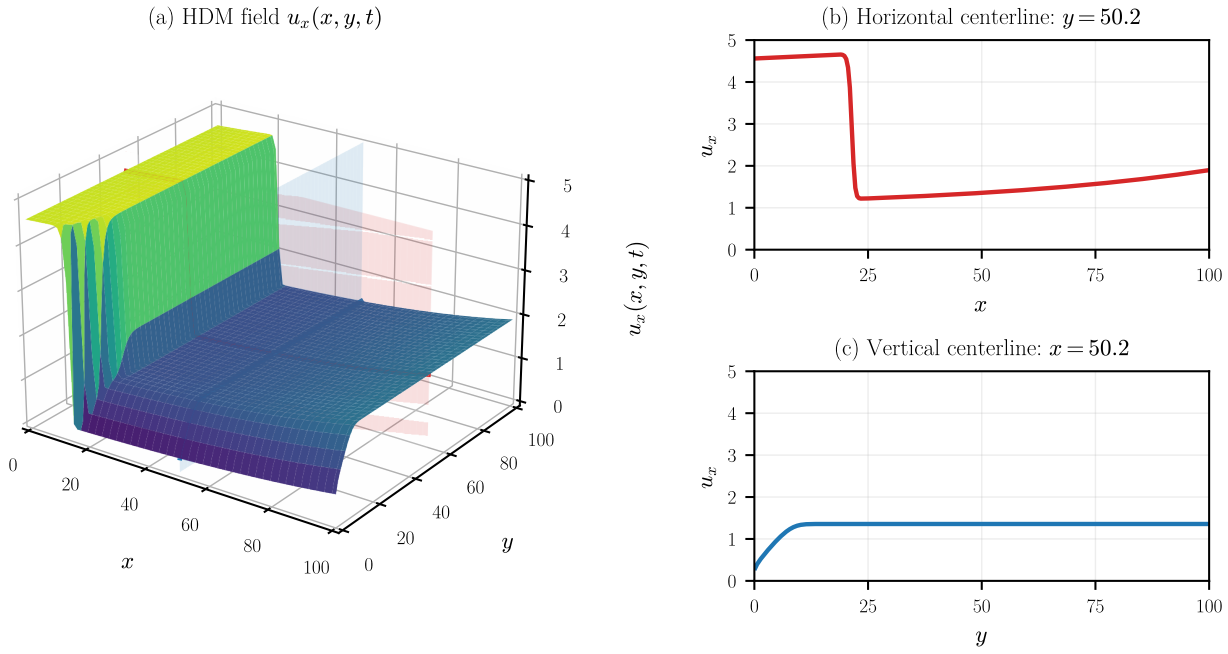


Figure 1: HDM solution at $\boldsymbol{\mu} = (4.56, 0.019)$ and $t = 7.5$: u_x over the computational domain and two centerline slices.

4.1 Reduced-order models and hyperreduction

The common affine trial space is constructed from HDM solution trajectories computed on the nine-point parameter grid

$$\{4.25, 4.875, 5.50\} \times \{0.015, 0.0225, 0.030\}. \tag{70}$$

Each trajectory contributes 501 states, including its initial state, yielding $N_s = 4\,509$ solution snapshots. Their mean defines the reference state $\boldsymbol{u}_{\text{ref}}$. Applying the POD construction of Section 1 to the centered snapshots gives a ROB with $n = 96$ modes, accounting for 99.95% of the sum of squared singular values. This ROB and reference state are used by both HPROMs.

To train the cubature rules, $N_c = 225$ state pairs are selected, with 25 pairs distributed over the simulation interval at each training parameter. Each pair contains states at consecutive time levels, as required by the implicit time discretization. Both states are projected onto the affine trial space using Equation (23), and the associated local residuals and test-basis blocks provide the projected entity contributions used in the fits.

Stacking these contributions as in Equation (25) gives the classical training matrix with 21 600 rows and 62 500 columns. For BM-ECSW, the same entries are rearranged into 96 modal matrices of size $225 \times 62\,500$, following Equation (47). Classical ECSW and BM-ECSW are trained using Algorithms 1 and 3, respectively, with relative tolerance $\tau = 10^{-5}$. Their sampled-mesh sizes are determined by attaining this tolerance.

Table 1 reports the resulting reduced dimension n, sampled-mesh size n_e, augmented-mesh size n_e^+, relative fitting error η, and estimated offline training time. The timings are obtained from one fresh run per method on 20 CPU threads and include assembly of the training matrix, mesh sampling, and weight fitting. The common costs of generating HDM trajectories and constructing the POD basis are excluded.

Table 1
Reduced-mesh sizes, relative fitting errors, and estimated offline training times for the 2D parametric inviscid Burgers problem.

Computational model	n	n_e	n_e^+	η	Offline training time (min)
HPROM (ECSW)	96	2 492	5 914	9.795×10^{-6}	112.56
HPROM (BM-ECSW)	96	300	868	8.359×10^{-6}	4.10

Both rules attain the prescribed training tolerance. BM-ECSW reduces the sampled-mesh size by a factor of 8.31 and also produces a smaller augmented mesh. The smaller meshes reduce the number of local residual and test-basis blocks evaluated at each online iteration. The measured offline training time is lower for BM-ECSW by a factor of 27.48.

The smaller BM mesh is accompanied by a larger set of scalar weights: 21 597 positive modal coefficients, compared with 2 492 weights for the classical rule. Figure 2 shows the sampled meshes together with the additional cells required by their residual stencils.

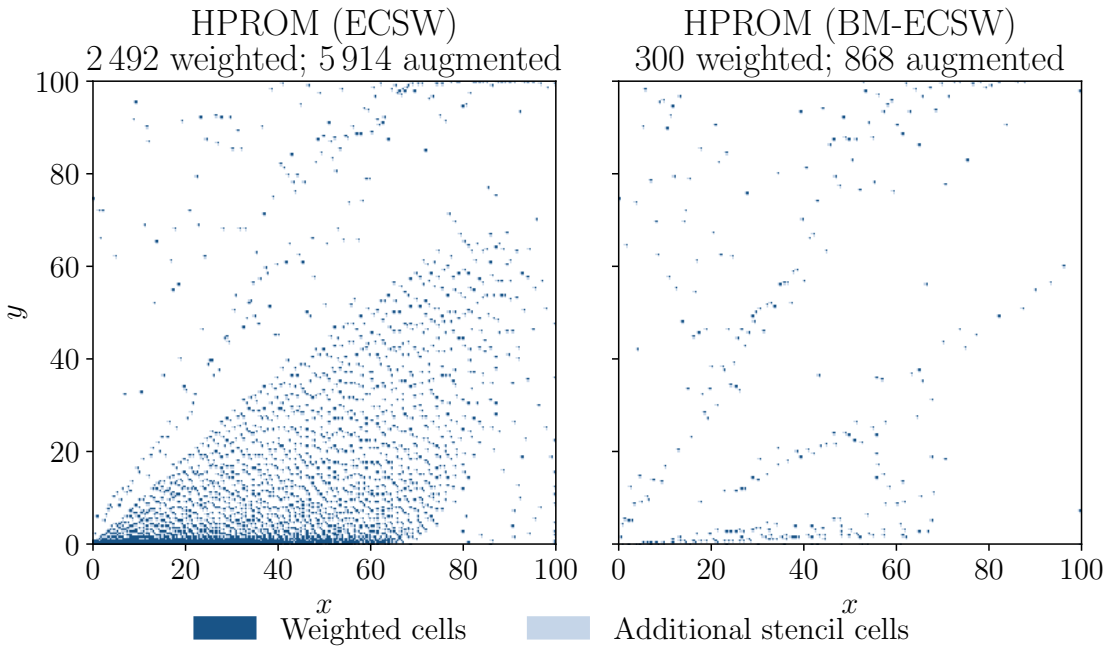


Figure 2: Reduced meshes for HPROM (ECSW) and HPROM (BM-ECSW). Dark blue marks weighted cells; light blue marks additional cells needed by the residual stencil.

4.2 Accuracy and online performance

The resulting HPROMs are evaluated at $\boldsymbol{\mu} = (4.56, 0.019)$, $(4.75, 0.020)$, and $(5.19, 0.026)$. These parameter points lie inside $\mathcal{D}$ and are absent from the nine-point training grid. The HDM supplies the reference solution and the timing baseline.

HPROM (ECSW) and HPROM (BM-ECSW) use corrections computed from the reduced systems in Algorithms 2 and 4, respectively. For these computations, both HPROMs use the previous reduced state as the initial guess at each time step and allow at most 20 nonlinear iterations. HPROM (ECSW) monitors the weighted sampled HDM residual norm, while HPROM (BM-ECSW) monitors the norm of its projected equations. Each solve uses a relative norm tolerance of 10^{-5} and a relative change threshold of 10^{-2} to detect a plateau. The classical solve applies full coordinate corrections; BM uses step control based on the decrease of its projected residual. These settings govern the online nonlinear solves and are separate from the cubature training tolerance.

Accuracy is measured over the complete solution trajectory using the global Euclidean relative error [1]

$$\mathbb{RE}_{2,\boldsymbol{u}}(\boldsymbol{\mu}) = \left[\frac{\sum_{m=0}^{N_t} \|\boldsymbol{u}^m_{\mathrm{HDM}}(\boldsymbol{\mu}) - \widetilde{\boldsymbol{u}}^m_{\mathrm{HPROM}}(\boldsymbol{\mu})\|_2^2}{\sum_{m=0}^{N_t} \|\boldsymbol{u}^m_{\mathrm{HDM}}(\boldsymbol{\mu})\|_2^2} \right]^{1/2}, \qquad N_t = 500. \tag{71}$$

This measure includes both velocity components over the complete trajectory and is reported in percent. The maximum error over the three queried parameters is defined as

$$\mathbb{RE}^{\max}_{2,\boldsymbol{u}} = \max_{\boldsymbol{\mu} \in \mathcal{D}_{\mathrm{test}}} \mathbb{RE}_{2,\boldsymbol{u}}(\boldsymbol{\mu}), \qquad \mathcal{D}_{\mathrm{test}} = \{(4.56, 0.019), (4.75, 0.020), (5.19, 0.026)\}. \tag{72}$$

All computations are performed in double-precision arithmetic on an Intel Core Ultra 7 265 processor with 20 CPU threads. Each HPROM is run three times at each test parameter, and

one HDM run per parameter supplies the reference time. The runs are executed sequentially, and the HDM nonlinear solver uses a relative residual tolerance of 10^{-12}. Online times include solver setup and all 500 time steps; offline training, error assessment, and postprocessing are excluded. The HPROM time at each parameter is the mean of its three repetitions. For the overall comparison, each HPROM time is averaged over nine runs and the HDM time over three runs. Each speedup factor is the corresponding HDM time divided by the mean HPROM time. Wall-clock times are reported in minutes.

Table 2
Online trajectory errors, wall-clock times, and speedup factors relative to the HDM at the three unsampled parameter points.

Parameter	Global Euclidean error (%)		Wall-clock time (min)			Speedup factor	
(μ_1, μ_2)	HPROM (ECSW)	HPROM (BM-ECSW)	HDM	HPROM (ECSW)	HPROM (BM-ECSW)	HPROM (ECSW)	HPROM (BM-ECSW)
$(4.56, 0.019)$	1.096	1.102	5.387	0.309	0.020	17.43	265.25
$(4.75, 0.020)$	1.001	1.006	5.388	0.314	0.020	17.17	268.52
$(5.19, 0.026)$	0.993	1.007	5.438	0.310	0.020	17.57	270.05

As shown in Table 2, both HPROMs yield global errors close to 1% at the three unsampled parameter points. The largest increase in trajectory error for HPROM (BM-ECSW) relative to HPROM (ECSW) at an individual parameter is 0.014 percentage points. Every BM trajectory completes all 500 steps with the projected-equation tolerance satisfied within the iteration budget. Using the smaller BM mesh therefore retains comparable trajectory accuracy for this set of queried parameters.

Figure 3 compares horizontal centerline profiles at $\boldsymbol{\mu} = (4.56, 0.019)$. Both HPROMs track the moving shock and closely follow the HDM profiles at the displayed times, with small oscillations near the discontinuity. These profiles illustrate the similar solution accuracy obtained with the two sampled meshes.

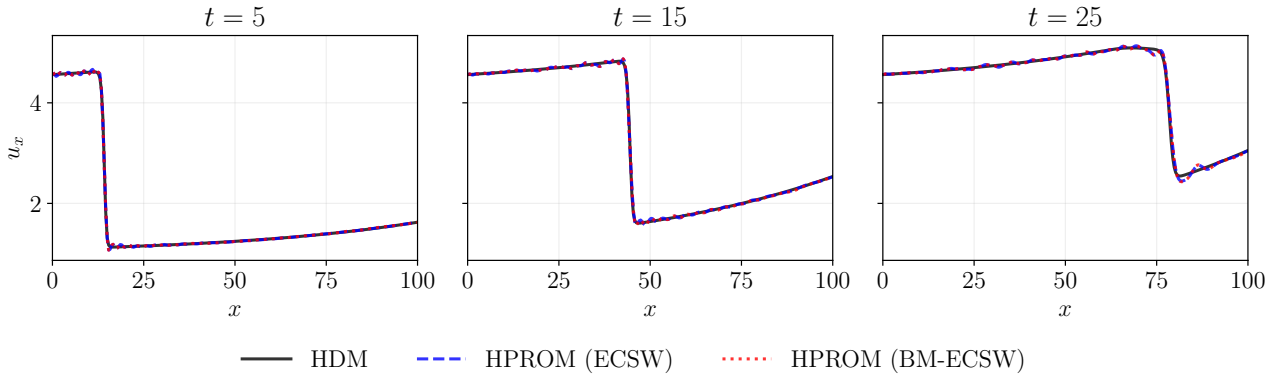


Figure 3: Horizontal centerline slices of u_x at $y = 50.2$, $\boldsymbol{\mu} = (4.56, 0.019)$, and $t = 5, 15, 25$. The three curves compare the HDM reference with HPROM (ECSW) and HPROM (BM-ECSW).

The overall performance is summarized in Table 3, using the maximum trajectory errors defined in Equation (72).

Table 3
Maximum trajectory errors, mean online times, and speedup factors relative to the HDM.

Computational model	n (N for HDM)	n_e (N_e for HDM)	n_e^+	$\mathbb{RE}_{2,\boldsymbol{u}}^{\max}$ (%)	Wall-clock time (min)	Speedup factor
HDM	125 000	62 500	62 500	–	5.404	–
HPROM (ECSW)	96	2 492	5 914	1.096	0.311	17.39
HPROM (BM-ECSW)	96	300	868	1.102	0.020	267.93

The overall speedup factors relative to the HDM are 17.39 for HPROM (ECSW) and 267.93 for HPROM (BM-ECSW). Individual HPROM times and sample standard deviations are reported in Appendix B.

The lower online cost of HPROM (BM-ECSW) is consistent with its smaller sampled and augmented meshes, which reduce the number of local residual and test-basis blocks evaluated during the nonlinear iterations. Both models use corrections assembled from linearized local residuals, but the measured solvers differ in their monitored norm and step control. The reported speedups therefore compare the complete online procedures at the stated accuracy.

5 Current assessment

BM-ECSW uses a shared reduced mesh and separate nonnegative modal fits against the full projected training targets. Its online Gauss–Newton-type update is assembled from local residuals and test-basis blocks on that mesh. The generalized coordinates are computed together.

For the two-dimensional parametric inviscid Burgers problem, the same 96-mode ROB, 225 training pairs, and 10^{-5} contribution-fit tolerance produce reduced meshes of 2 492 cells for classical ECSW and 300 cells for BM-ECSW. At three unsampled parameter points, the global errors remain close to 1%. The measured mean-time speedup factors relative to the HDM are 17.39 for HPROM (ECSW) and 267.93 for HPROM (BM-ECSW). BM-ECSW also reduces the measured offline training time by a factor of 27.48.

Mode-dependent weights enlarge the set of cubature fits available on a selected mesh. Their effect on gradient structure follows from the general compatibility condition in Appendix A.1. For LSPG, the BM projected equations are not guaranteed to be derivatives of one common sampled residual objective (Appendix A.2). The online correction matrix can also be nonsymmetric and have a quadratic form that takes negative values (Appendix A.2.3). In conservative structural dynamics, the inherited mechanical Lagrangian form and the associated energy-conservation construction are likewise not guaranteed (Appendix A.3). The projected-residual training tolerance does not impose these structural conditions. The exact-Jacobian checks in Appendix A.4.1 confirm loss of a common objective in both Burgers BM rules, including the accurate rule with $N_c = 225$.

The present results establish trajectory accuracy and computational cost for the stated Burgers configuration. Appendix A.4.1 examines the earlier 90-pair rule and shows why a small training error alone does not ensure accurate online equations. Behavior outside the sampled parameter range and stability in structural applications remain to be assessed.

A Common objectives, symmetry, and energy under modal weighting

Modal weights can reduce the sampled mesh, but they do not guarantee that the weighted vector remains the gradient of one common scalar function. In LSPG, the question is whether the weighted projected residual remains the gradient of one common scalar least-squares objective; in conservative structural dynamics, whether the weighted internal-force vector remains the gradient of one common scalar potential energy. The general condition is derived first, then applied to these two settings and the corresponding correction and stiffness matrices. Analytical examples establish possible losses of structure and stability; AERO-F and Burgers computations examine the behavior of trained rules.

A.1 Preserving the gradient of one common scalar function

Consider twice continuously differentiable local scalar functions $\phi_e(\boldsymbol{q})$, with $\boldsymbol{q} \in \mathbb{R}^n$. The question is whether each assembled vector is the gradient of one common scalar function of all the coordinates. In the weighted constructions, the sampled set and weights remain fixed.

The unweighted vector is

$$\boldsymbol{F} = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e. \tag{73}$$

If one common twice continuously differentiable scalar function Φ satisfies $\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi$, then $F_i = \partial\Phi/\partial q_i$. Differentiating component i with respect to q_j and component j with respect to q_i must therefore give the same mixed second derivative:

$$\frac{\partial F_i}{\partial q_j} - \frac{\partial F_j}{\partial q_i} = \frac{\partial^2\Phi}{\partial q_j\partial q_i} - \frac{\partial^2\Phi}{\partial q_i\partial q_j} = 0, \qquad i,j = 1,\dots,n. \tag{74}$$

This is a necessary compatibility condition for any vector to be the gradient of one common smooth scalar function. For the vector in Equation (73), $F_i = \sum_{e\in\mathcal{E}} \partial\phi_e/\partial q_i$. Differentiating its components gives

$$\frac{\partial F_i}{\partial q_j} = \sum_{e\in\mathcal{E}} \frac{\partial^2\phi_e}{\partial q_j\partial q_i}, \qquad \frac{\partial F_j}{\partial q_i} = \sum_{e\in\mathcal{E}} \frac{\partial^2\phi_e}{\partial q_i\partial q_j}.$$

Their difference is therefore

$$\frac{\partial F_i}{\partial q_j} - \frac{\partial F_j}{\partial q_i} = \sum_{e\in\mathcal{E}} \underbrace{\left[\frac{\partial^2\phi_e}{\partial q_j\partial q_i} - \frac{\partial^2\phi_e}{\partial q_i\partial q_j}\right]}_{=0} = 0.$$

Each local pair agrees because ϕ_e is twice continuously differentiable. Thus $\boldsymbol{F}$ satisfies Equation (74). For this vector, a common scalar function can be identified explicitly. Defining this function and differentiating gives

$$\Phi(\boldsymbol{q}) = \sum_{e\in\mathcal{E}} \phi_e(\boldsymbol{q}),$$

$$\nabla_{\boldsymbol{q}}\Phi = \nabla_{\boldsymbol{q}}\left[\sum_{e\in\mathcal{E}} \phi_e\right] = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e = \boldsymbol{F}.$$

On a fixed sampled set $\widetilde{\mathcal{E}} \subseteq \mathcal{E}$, fixed scalar weights, as in classical ECSW, give the vector

$$\boldsymbol{F}_{\mathrm{ECSW}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e.$$

Different entities can have different weights, while each ξ_e multiplies every partial derivative of ϕ_e. To apply the same test in Equation (74), write component i as

$$F_{\mathrm{ECSW},i} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \frac{\partial \phi_e}{\partial q_i}.$$

The weights are fixed, so differentiating the two components gives

$$\begin{aligned} \frac{\partial F_{\mathrm{ECSW},i}}{\partial q_j} &= \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \frac{\partial^2 \phi_e}{\partial q_j \partial q_i}, \\ \frac{\partial F_{\mathrm{ECSW},j}}{\partial q_i} &= \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}. \end{aligned}$$

Subtracting yields

$$\frac{\partial F_{\mathrm{ECSW},i}}{\partial q_j} - \frac{\partial F_{\mathrm{ECSW},j}}{\partial q_i} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \underbrace{\left[\frac{\partial^2 \phi_e}{\partial q_j \partial q_i} - \frac{\partial^2 \phi_e}{\partial q_i \partial q_j} \right]}_{=0} = 0.$$

The two local mixed derivatives agree and share the same scalar weight ξ_e. Every entity therefore contributes zero, so $\boldsymbol{F}_{\mathrm{ECSW}}$ satisfies Equation (74). The common scalar function can now be identified and its gradient verified:

$$\begin{aligned} \Phi_{\mathrm{ECSW}}(\boldsymbol{q}) &= \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \phi_e(\boldsymbol{q}), \\ \nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}} &= \nabla_{\boldsymbol{q}} \left[\sum_{e \in \widetilde{\mathcal{E}}} \xi_e \phi_e \right] = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{F}_{\mathrm{ECSW}}. \end{aligned} \tag{75}$$

Component-dependent weighting, as in BM-ECSW, instead constructs

$$\boldsymbol{F}_{\mathrm{BM}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e, \qquad \boldsymbol{D}_e = \mathrm{diag}(\xi_{e,1}, \ldots, \xi_{e,n}). \tag{76}$$

The matrices are fixed, but different partial derivatives of one function can receive different weights. Component i is

$$F_{\mathrm{BM},i} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \phi_e}{\partial q_i}. \tag{77}$$

If a common scalar function Φ_{BM} exists such that $\boldsymbol{F}_{\mathrm{BM}} = \nabla_{\boldsymbol{q}} \Phi_{\mathrm{BM}}$, this vector must satisfy the compatibility condition in Equation (74). To check this condition, differentiate its components:

$$\begin{aligned} \frac{\partial F_{\mathrm{BM},i}}{\partial q_j} &= \sum_{e \in \widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial^2 \phi_e}{\partial q_j \partial q_i}, \\ \frac{\partial F_{\mathrm{BM},j}}{\partial q_i} &= \sum_{e \in \widetilde{\mathcal{E}}} \xi_{e,j} \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}. \end{aligned} \tag{78}$$

The local mixed derivatives agree, but the weights multiplying them can differ. Subtracting yields

$$\frac{\partial F_{\mathrm{BM},i}}{\partial q_j} - \frac{\partial F_{\mathrm{BM},j}}{\partial q_i} = \underbrace{\sum_{e\in\tilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}}_{\text{not necessarily zero}} . \tag{79}$$

For a common Φ_{BM} to generate the BM vector, this sum must vanish for every pair i, j. A nonzero sum at a point excludes every such smooth function locally. If all these sums vanish throughout an open simply connected domain, such a function exists there. Equal component weights, $\xi_{e,i} = \xi_{e,j} = \xi_e$, make every term zero. Unequal weights can also satisfy the condition if local mixed derivatives vanish or contributions cancel.

One fixed scalar weight per entity preserves the gradient of one common scalar function; separate weights for its reduced components do not guarantee this property.

In LSPG, loss of gradient structure removes the common least-squares interpretation and the classical descent argument based on that objective (Section A.2).

In structural dynamics, loss of gradient structure excludes the common potential required for the inherited Lagrangian and energy-conservation construction with the same reduced kinetic energy (Section A.3).

In practice, the LSPG iteration may stall or oscillate, require more damping or iterations, or fail to reach the nonlinear tolerance. In structural dynamics, modal weighting can produce growing displacements despite a stable classical response (Section A.3.2); conservation of the inherited mechanical energy is no longer guaranteed. These are possible outcomes, and the structural losses alone do not establish the cause of a particular failure.

A.2 LSPG objectives, online corrections, and matrix properties

The general construction of Section A.1 is applied to LSPG by defining the local scalar functions as

$$\phi_e(\boldsymbol{q}) = \frac{1}{2}\|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2 = \frac{1}{2}\boldsymbol{r}_e^T\boldsymbol{r}_e.$$

The blocks $\boldsymbol{r}_e$ partition the rows of the assembled HDM residual evaluated at the reconstructed state. With $\boldsymbol{W}_e = \partial \boldsymbol{r}_e / \partial \boldsymbol{q}$, differentiating each local function gives

$$\nabla_{\boldsymbol{q}} \phi_e = \frac{1}{2}\left[\boldsymbol{W}_e^T\boldsymbol{r}_e + [\boldsymbol{r}_e^T\boldsymbol{W}_e]^T\right] = \boldsymbol{W}_e^T\boldsymbol{r}_e. \tag{80}$$

The two terms in brackets are equal and cancel the factor $1/2$. Assembling these local gradients gives

$$\boldsymbol{F} = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e = \sum_{e\in\mathcal{E}} \boldsymbol{W}_e^T\boldsymbol{r}_e = \boldsymbol{r}_n.$$

With these local functions, the vector $\boldsymbol{F}$ of Section A.1 is the full LSPG projected residual defined in Equation (9). The mixed derivatives of each ϕ_e agree, so this vector satisfies the compatibility condition in Equation (74). Its common scalar function is the LSPG objective in Equation (7):

$$\Phi(\boldsymbol{q}) = \sum_{e\in\mathcal{E}} \phi_e(\boldsymbol{q}) = \frac{1}{2}\|\boldsymbol{r}(\boldsymbol{q})\|_2^2. \tag{81}$$

Differentiating verifies that its gradient reproduces the complete projected vector:

$$\nabla_{\boldsymbol{q}}\Phi = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e = \sum_{e\in\mathcal{E}} \boldsymbol{W}_e^T \boldsymbol{r}_e = \boldsymbol{r}_n. \tag{82}$$

Setting $\boldsymbol{r}_n = \boldsymbol{0}$ therefore imposes stationarity of this one scalar function.

Classical ECSW. Classical ECSW approximates $\boldsymbol{r}_n$ using a sampled set and one fixed positive weight ξ_e per entity. With the local functions defined above, the scalar-weighted construction of Section A.1 gives

$$\boldsymbol{F}_{\mathrm{ECSW}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e. \tag{83}$$

Weights may differ between entities, but each fixed ξ_e multiplies all reduced components of one entity. The same weight therefore multiplies both local mixed derivatives, so their difference vanishes in the test of Equation (74). For these squared-residual functions, Equation (75) identifies the common sampled least-squares objective

$$\Phi_{\mathrm{ECSW}}(\boldsymbol{q}) = \frac{1}{2}\sum_{e\in\widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2 = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e. \tag{84}$$

With the weights fixed, differentiation gives

$$\nabla_{\boldsymbol{q}}\Phi_{\mathrm{ECSW}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}}\phi_e = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}. \tag{85}$$

Thus $\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \boldsymbol{0}$ imposes stationarity of one common sampled least-squares objective. This objective generally differs from the full LSPG objective Φ.

BM-ECSW. BM-ECSW assigns separate weights to the reduced components of each entity's contribution. With these local functions, the component-weighted construction of Section A.1 gives

$$\boldsymbol{F}_{\mathrm{BM}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{r}_e. \tag{86}$$

Using Equation (80), component i is

$$\widetilde{r}_{n,i}^{\mathrm{BM}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \boldsymbol{W}_e(:,i)^T \boldsymbol{r}_e = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \phi_e}{\partial q_i}. \tag{87}$$

For a common scalar objective Φ_{BM} to satisfy $\nabla_{\boldsymbol{q}}\Phi_{\mathrm{BM}} = \boldsymbol{F}_{\mathrm{BM}}$, the two mixed derivatives must agree, as required by Equation (74). With the weights fixed, differentiating Equation (87) gives

$$\frac{\partial \widetilde{r}_{n,i}^{\mathrm{BM}}}{\partial q_j} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial^2 \phi_e}{\partial q_j \partial q_i},$$
$$\frac{\partial \widetilde{r}_{n,j}^{\mathrm{BM}}}{\partial q_i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,j} \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}.$$

The local mixed derivatives agree, but their weights can differ. Subtracting yields

$$\frac{\partial \widetilde{r}_{n,i}^{\mathrm{BM}}}{\partial q_j} - \frac{\partial \widetilde{r}_{n,j}^{\mathrm{BM}}}{\partial q_i} = \underbrace{\sum_{e \in \widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}}_{\text{not necessarily zero}}.$$

This is Equation (79) applied to the BM-ECSW projected vector $\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}$. A nonzero difference at a point excludes a common smooth objective Φ_{BM} locally. BM therefore does not guarantee a common objective, although its online problem remains $\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}) = \boldsymbol{0}$.

A.2.1 Numerical example of the mixed-derivative test

Consider three scalar residual blocks depending on two reduced coordinates:

$$\boldsymbol{r}(\boldsymbol{q}) = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} q_1 \\ q_2 \\ q_1 + q_2^2 - 3 \end{pmatrix}. \tag{88}$$

Each residual block is scalar, so its test-basis row is

$$\boldsymbol{W}_1 = (1, 0), \qquad \boldsymbol{W}_2 = (0, 1), \qquad \boldsymbol{W}_3 = (1, 2q_2).$$

The two rules use the same residual blocks, test-basis rows, and three entities. Only the weight assigned to entity 3 in the second reduced equation changes. Equation (74) gives the necessary mixed-derivative condition for each assembled vector to be the gradient of one common scalar objective. Rows of $\boldsymbol{\Xi}$ identify entities, and columns identify reduced equations.

Classical ECSW

The entity weights are

$$\boldsymbol{\xi} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}. \tag{89a}$$

The classical ECSW approximation is

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \sum_{e=1}^{3} \xi_e \boldsymbol{W}_e^T r_e. \tag{90a}$$

Assembling the weighted contributions gives

$$\begin{aligned} \widetilde{r}_{n,1}^{\mathrm{ECSW}} &= 2r_1 + 2r_3 \\ &= 2q_1 + 2r_3, \\ \widetilde{r}_{n,2}^{\mathrm{ECSW}} &= 3r_2 + 4q_2 r_3 \\ &= 3q_2 + 4q_2 r_3. \end{aligned} \tag{91a}$$

BM-ECSW

The modal weights are

$$\boldsymbol{\Xi} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \\ 2 & 1 \end{pmatrix}. \tag{89b}$$

The BM-ECSW approximation is

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = \sum_{e=1}^{3} \boldsymbol{D}_e \boldsymbol{W}_e^T r_e. \tag{90b}$$

Assembling the weighted contributions gives

$$\begin{aligned} \widetilde{r}_{n,1}^{\mathrm{BM}} &= 2r_1 + 2r_3 \\ &= 2q_1 + 2r_3, \\ \widetilde{r}_{n,2}^{\mathrm{BM}} &= 3r_2 + 2q_2 r_3 \\ &= 3q_2 + 2q_2 r_3. \end{aligned} \tag{91b}$$

To apply the compatibility test to these assembled vectors, differentiate the first component with respect to q_2 and the second with respect to q_1:

The mixed derivatives are

$$\frac{\partial \widetilde{r}_{n,1}^{\mathrm{ECSW}}}{\partial q_2} = 2\frac{\partial r_3}{\partial q_2} = 4q_2,$$
$$\frac{\partial \widetilde{r}_{n,2}^{\mathrm{ECSW}}}{\partial q_1} = 4q_2\frac{\partial r_3}{\partial q_1} = 4q_2. \tag{92a}$$

The mixed derivatives agree for all $\boldsymbol{q}$. The common scalar objective is

$$\Phi_{\mathrm{ECSW}} = \frac{1}{2}\left[2q_1^2 + 3q_2^2 + 2r_3^2\right].$$

Its gradient reproduces both classical components in Equation (91).

The mixed derivatives are

$$\frac{\partial \widetilde{r}_{n,1}^{\mathrm{BM}}}{\partial q_2} = 2\frac{\partial r_3}{\partial q_2} = 4q_2,$$
$$\frac{\partial \widetilde{r}_{n,2}^{\mathrm{BM}}}{\partial q_1} = 2q_2\frac{\partial r_3}{\partial q_1} = 2q_2. \tag{92b}$$

The mixed derivatives differ by $2q_2$, violating Equation (74) whenever $q_2 \neq 0$. Entities 1 and 2 contribute zero to this difference; entity 3 contributes $2q_2$. Hence no smooth scalar function has this BM vector as its gradient near such a point.

The common classical objective supplies the least-squares model for its online correction. The BM correction is formed from the weighted projected equations, without assuming a common objective.

A.2.2 Online correction: from Gauss–Newton to a Gauss–Newton-type update

At the current iterate, both methods hold the test-basis blocks fixed while computing the increment and approximate each local residual by $\boldsymbol{r}_e(\boldsymbol{q} + \Delta\boldsymbol{q}) \approx \boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}$. For classical ECSW, the Gauss–Newton increment minimizes the quadratic model obtained by substituting these linearized residuals into the sampled objective:

$$\min_{\Delta\boldsymbol{q}\in\mathbb{R}^n} \frac{1}{2}\sum_{e\in\widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}\|_2^2. \tag{93}$$

Differentiating with respect to the increment gives its normal equations, the Gauss–Newton system in Equation (42). For nonnegative weights, this quadratic model is convex, so any solution of its normal equations minimizes it over the complete increment vector.

For BM, applying the modal weights to the projected linearized contributions and setting their sum to zero gives:

$$\sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e\boldsymbol{W}_e^T[\boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}] = \boldsymbol{0}. \tag{94}$$

Expanding gives the correction system

$$\left[\sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e\boldsymbol{W}_e^T\boldsymbol{W}_e\right]\Delta\boldsymbol{q} = -\sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e\boldsymbol{W}_e^T\boldsymbol{r}_e. \tag{95}$$

This is the update in Equation (66). It uses a Gauss–Newton-type approximation to the Jacobian of the BM projected vector: terms involving local residuals multiplied by their second derivatives are omitted. The increment therefore requires only local residuals and their first derivatives.

Both methods compute one coupled increment. **The classical Gauss–Newton correction minimizes one common quadratic error; the BM correction solves weighted linearized equations without that guarantee.**

A.2.3 Symmetry and quadratic form

Section A.1 tests whether the assembled vector is the gradient of one common scalar function using its exact Jacobian. The online corrections use approximate Jacobians. Their symmetry and quadratic forms are examined here:

$$\begin{aligned} \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} &= \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{W}_e, \\ \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} &= \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e. \end{aligned} \tag{96}$$

Each local product $\boldsymbol{W}_e^T\boldsymbol{W}_e$ is symmetric. The classical scalar weights preserve that symmetry:

$$\big[\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}\big]^T = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \big[\boldsymbol{W}_e^T\boldsymbol{W}_e\big]^T = \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}.$$

Moreover, nonnegative weights give, for any $\boldsymbol{z}\in\mathbb{R}^n$,

$$\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}\boldsymbol{z} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e\boldsymbol{z}]^T[\boldsymbol{W}_e\boldsymbol{z}] = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{W}_e\boldsymbol{z}\|_2^2 \geq 0. \tag{97}$$

The classical correction matrix is therefore symmetric positive semidefinite [7].

The BM matrix instead multiplies row i of each local product by $\xi_{e,i}$. The difference between its crossed entries is the first-derivative approximation of Equation (79): the local mixed derivative is replaced by $\boldsymbol{W}_e(:,i)^T\boldsymbol{W}_e(:,j)$, giving

$$\big[\widehat{J}_n^{\mathrm{BM}}\big]_{ij} - \big[\widehat{J}_n^{\mathrm{BM}}\big]_{ji} = \sum_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}]\boldsymbol{W}_e(:,i)^T\boldsymbol{W}_e(:,j). \tag{98}$$

The local inner product is unchanged when i and j are interchanged, but the weights can differ. The matrix can consequently be nonsymmetric. Its quadratic form is

$$\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}\boldsymbol{z} = \sum_{e\in\widetilde{\mathcal{E}}} [\boldsymbol{W}_e\boldsymbol{D}_e\boldsymbol{z}]^T[\boldsymbol{W}_e\boldsymbol{z}]. \tag{99}$$

Unlike the classical squared norms, these dot products can be negative. **Nonnegative modal weights do not guarantee a symmetric correction matrix or a nonnegative quadratic form.** Each local product $\boldsymbol{W}_e^T\boldsymbol{W}_e$ remains symmetric positive semidefinite, and setting $\boldsymbol{D}_e = \xi_e\boldsymbol{I}_n$ recovers the classical matrix and both guarantees.

The quadratic form $\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z}$ is a scalar and therefore equals its transpose:

$$\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z} = \big[\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z}\big]^T = \boldsymbol{z}^T\widehat{\boldsymbol{J}}_n^T\boldsymbol{z}.$$

Averaging these equal expressions leaves their value unchanged:

$$\begin{aligned} \boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z} &= \frac{1}{2}\big[\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z} + \boldsymbol{z}^T\widehat{\boldsymbol{J}}_n^T\boldsymbol{z}\big] \\ &= \boldsymbol{z}^T\,\mathrm{sym}(\widehat{\boldsymbol{J}}_n)\boldsymbol{z}, \qquad \mathrm{sym}(\widehat{\boldsymbol{J}}_n) = \frac{1}{2}\big[\widehat{\boldsymbol{J}}_n + \widehat{\boldsymbol{J}}_n^T\big]. \end{aligned} \tag{100}$$

The original matrix and its symmetric part therefore give the same quadratic form for every $\boldsymbol{z}$. Its sign can be determined from the symmetric part. The classical matrix equals its symmetric part, and Equation (97) establishes that its quadratic form is nonnegative. For either matrix, if $\boldsymbol{v}$ is an eigenvector of its symmetric part with eigenvalue λ, then

$$\boldsymbol{v}^T \widehat{\boldsymbol{J}}_n \boldsymbol{v} = \boldsymbol{v}^T \operatorname{sym}(\widehat{\boldsymbol{J}}_n)\boldsymbol{v} = \lambda \|\boldsymbol{v}\|_2^2.$$

Thus a negative eigenvalue of the BM symmetric part identifies a direction with a negative quadratic form.

A.2.4 A matrix example with positive eigenvalues

In the auxiliary linear model $\dot{\boldsymbol{z}} = -\widehat{\boldsymbol{J}}_n \boldsymbol{z}$, the matrix is held fixed and $\boldsymbol{z}$ represents a deviation from zero. Its size is measured by $\|\boldsymbol{z}\|_2$. Along a solution, differentiating the squared norm and substituting the model equation gives

$$\begin{aligned} \frac{d}{dt}\frac{1}{2}\|\boldsymbol{z}\|_2^2 &= \boldsymbol{z}^T \dot{\boldsymbol{z}} \\ &= -\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n \boldsymbol{z} = -\boldsymbol{z}^T \operatorname{sym}(\widehat{\boldsymbol{J}}_n)\boldsymbol{z}. \end{aligned} \tag{101}$$

The minus sign in Equation (101) gives two consequences:

- A positive quadratic form makes the perturbation norm decrease.
- A negative quadratic form makes the perturbation norm initially increase.

A positive definite symmetric part gives a decreasing norm for every nonzero perturbation. An indefinite symmetric part permits initial growth, even when every eigenvalue of the full matrix has a positive real part. The HPROM trajectories are examined in Section A.4.

Consider three scalar residual blocks with fixed test-basis rows

$$\boldsymbol{W}_1 = (1, 0), \qquad \boldsymbol{W}_2 = (0, 1), \qquad \boldsymbol{W}_3 = (1, 1).$$

Both rules use all three entities. The BM rule assigns entity 3 a zero weight in the second reduced equation, which nonnegativity permits.

Classical ECSW

The entity weights $\boldsymbol{\xi} = (1, 1, 4)^T$ give

$$\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} = \sum_{e=1}^{3} \xi_e \boldsymbol{W}_e^T \boldsymbol{W}_e = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}. \tag{102a}$$

The matrix is symmetric, with eigenvalues 1 and 9, so it is positive definite. For $\boldsymbol{z} = (1, -2)^T$,

$$\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} \boldsymbol{z} = 9,$$

and Equation (101) gives an initial decrease of $\|\boldsymbol{z}\|_2$.

BM-ECSW

The modal weights $\boldsymbol{\Xi} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 8 & 0 \end{pmatrix}$ give

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}} = \sum_{e=1}^{3} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e = \begin{pmatrix} 9 & 8 \\ 0 & 1 \end{pmatrix}. \tag{102b}$$

The matrix is triangular, with eigenvalues 9 and 1. Its symmetric part,

$$\operatorname{sym}(\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}) = \begin{pmatrix} 9 & 4 \\ 4 & 1 \end{pmatrix},$$

has determinant -7 and eigenvalues $5 \pm 4\sqrt{2}$. For $\boldsymbol{z} = (1, -2)^T$, $\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} \boldsymbol{z} = -3$, and $\|\boldsymbol{z}\|_2$ initially grows.

Both matrices have eigenvalues 1 and 9, so every perturbation eventually decays in this fixed linear model. The classical model decreases every perturbation's norm from the start. **The matrix example demonstrates temporary perturbation growth in a stable auxiliary model.**

The structural example in Section A.3.2 demonstrates a distinct possibility: a negative stiffness eigenvalue and an unstable mode despite positive weights.

A.3 Lagrangian structure and stability under modal weighting

The energy-conservation analysis of ECSW by Farhat et al. [5] motivates applying Section A.1 to conservative structural dynamics, with element potential energies as the local scalar functions. Their gradients generate the internal vector, and retaining one common potential is required to preserve the mechanical Lagrangian form with the same reduced kinetic energy. For a structural HDM with a constant symmetric positive definite mass matrix $\boldsymbol{M}$ and no external loading, the equations of motion are

$$\boldsymbol{M}\ddot{\boldsymbol{u}} + \nabla_{\boldsymbol{u}}\mathcal{V}^{\text{int}}(\boldsymbol{u}) = \boldsymbol{0}, \qquad \mathcal{V}^{\text{int}}(\boldsymbol{u}) = \sum_{e\in\mathcal{E}} \mathcal{V}_e^{\text{int}}(\boldsymbol{L}_e\boldsymbol{u}). \tag{103}$$

Here $\boldsymbol{L}_e$ extracts the displacement entries of element e, and the element potentials are twice continuously differentiable. The restoring force is $-\nabla_{\boldsymbol{u}}\mathcal{V}^{\text{int}}$.

Substituting $\boldsymbol{u} = \boldsymbol{V}\boldsymbol{q}$ with the fixed basis $\boldsymbol{V}$ and premultiplying by $\boldsymbol{V}^T$ gives the Galerkin structural model. Its reduced mass matrix and the local scalar functions of Section A.1 are

$$\boldsymbol{M}_r = \boldsymbol{V}^T\boldsymbol{M}\boldsymbol{V}, \qquad \phi_e(\boldsymbol{q}) = \mathcal{V}_{e,r}^{\text{int}}(\boldsymbol{q}) = \mathcal{V}_e^{\text{int}}(\boldsymbol{L}_e\boldsymbol{V}\boldsymbol{q}).$$

With these local functions, the vector $\boldsymbol{F}$ of Section A.1 is the reduced internal vector:

$$\boldsymbol{F} = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e = \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\mathcal{V}_{e,r}^{\text{int}}.$$

The mixed derivatives of each potential agree, so this vector satisfies Equation (74). Its common scalar function is the total reduced potential energy:

$$\Phi(\boldsymbol{q}) = \sum_{e\in\mathcal{E}} \phi_e(\boldsymbol{q}) = \sum_{e\in\mathcal{E}} \mathcal{V}_{e,r}^{\text{int}}(\boldsymbol{q}), \qquad \nabla_{\boldsymbol{q}}\Phi = \boldsymbol{F}.$$

The reduced kinetic energy and mechanical Lagrangian are then

$$\mathcal{T}_r(\dot{\boldsymbol{q}}) = \frac{1}{2}\dot{\boldsymbol{q}}^T\boldsymbol{M}_r\dot{\boldsymbol{q}}, \qquad \mathcal{L}_r(\boldsymbol{q},\dot{\boldsymbol{q}}) = \mathcal{T}_r(\dot{\boldsymbol{q}}) - \sum_{e\in\mathcal{E}} \mathcal{V}_{e,r}^{\text{int}}(\boldsymbol{q}). \tag{104}$$

Here $\mathcal{L}_r = \mathcal{T}_r - \Phi$. Hamilton's principle requires the time integral of this Lagrangian to be stationary under variations with fixed endpoints. Its Euler–Lagrange equations recover the Galerkin equations of motion, $\boldsymbol{M}_r\ddot{\boldsymbol{q}} + \boldsymbol{F} = \boldsymbol{0}$:

$$\frac{d}{dt}\left[\frac{\partial\mathcal{L}_r}{\partial\dot{\boldsymbol{q}}}\right] - \frac{\partial\mathcal{L}_r}{\partial\boldsymbol{q}} = \boldsymbol{M}_r\ddot{\boldsymbol{q}} + \sum_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\mathcal{V}_{e,r}^{\text{int}} = \boldsymbol{0}. \tag{105}$$

Classical ECSW uses the scalar-weighted vector of Section A.1:

$$\boldsymbol{F}_{\mathrm{ECSW}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \mathcal{V}^{\mathrm{int}}_{e,r}.$$

The fixed scalar weights preserve compatibility. Equation (75) therefore identifies one common sampled potential and the associated mechanical Lagrangian, with the same kinetic energy:

$$\Phi_{\mathrm{ECSW}} = \widetilde{\mathcal{V}}^{\mathrm{int}}_r = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \mathcal{V}^{\mathrm{int}}_{e,r}, \qquad \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r = \mathcal{T}_r - \widetilde{\mathcal{V}}^{\mathrm{int}}_r. \tag{106}$$

The gradient identity is $\nabla_{\boldsymbol{q}}\Phi_{\mathrm{ECSW}} = \boldsymbol{F}_{\mathrm{ECSW}}$. Since the mass matrix and weights are fixed, differentiating the Lagrangian gives

$$\frac{\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{M}_r \dot{\boldsymbol{q}}, \qquad \frac{\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r}{\partial \boldsymbol{q}} = -\sum_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \mathcal{V}^{\mathrm{int}}_{e,r}.$$

Substitution into the Euler–Lagrange equations yields

$$\frac{d}{dt}\left[\frac{\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r}{\partial \dot{\boldsymbol{q}}}\right] - \frac{\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r}{\partial \boldsymbol{q}} = \boldsymbol{M}_r \ddot{\boldsymbol{q}} + \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \mathcal{V}^{\mathrm{int}}_{e,r} = \boldsymbol{0}. \tag{107}$$

These are exactly the classical ECSW equations of motion. Thus sampling changes the potential while preserving the mechanical Lagrangian form. The associated energy is defined from the Lagrangian in Equation (106). Substituting its velocity derivative, $\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r / \partial \dot{\boldsymbol{q}} = \boldsymbol{M}_r \dot{\boldsymbol{q}}$, gives

$$\begin{aligned}
\widetilde{\mathcal{H}}_r &= \dot{\boldsymbol{q}}^T \frac{\partial \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r}{\partial \dot{\boldsymbol{q}}} - \widetilde{\mathcal{L}}^{\mathrm{ECSW}}_r \\
&= \dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}} - \left[\frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}} - \widetilde{\mathcal{V}}^{\mathrm{int}}_r\right] \\
&= \mathcal{T}_r + \widetilde{\mathcal{V}}^{\mathrm{int}}_r, \\
\frac{d\widetilde{\mathcal{H}}_r}{dt} &= \dot{\boldsymbol{q}}^T \left[\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}^{\mathrm{int}}_r\right] = 0.
\end{aligned} \tag{108}$$

The final equality follows from the equations of motion in Equation (107). This energy is conserved in continuous time; conservation after time discretization depends on the integrator.

BM-ECSW uses the component-weighted vector of Section A.1 as its internal vector:

$$\begin{aligned}
\boldsymbol{F}_{\mathrm{BM}} &= \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e = \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \mathcal{V}^{\mathrm{int}}_{e,r}, \\
\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \boldsymbol{F}_{\mathrm{BM}} &= \boldsymbol{0}.
\end{aligned} \tag{109}$$

The question from Section A.1 is now whether a common scalar function Φ_{BM} exists with $\nabla_{\boldsymbol{q}}\Phi_{\mathrm{BM}} = \boldsymbol{F}_{\mathrm{BM}}$. Such a function would be the potential $\Phi_{\mathrm{BM}} = \widetilde{\mathcal{V}}^{\mathrm{int,BM}}_r$ in the mechanical Lagrangian with the same kinetic energy:

$$\widetilde{\mathcal{L}}^{\mathrm{BM}}_r(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \mathcal{T}_r(\dot{\boldsymbol{q}}) - \widetilde{\mathcal{V}}^{\mathrm{int,BM}}_r(\boldsymbol{q}). \tag{110}$$

Its Euler–Lagrange equations reproduce Equation (109) if and only if the potential satisfies

$$\nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}_r^{\mathrm{int,BM}} = \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \mathcal{V}_{e,r}^{\mathrm{int}} = \boldsymbol{F}_{\mathrm{BM}}, \qquad \boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}_r^{\mathrm{int,BM}} = \boldsymbol{0}. \tag{111}$$

The first equality is a condition on the unknown potential. For any smooth candidate potential, the associated energy is

$$\widetilde{\mathcal{H}}_r^{\mathrm{BM}} = \mathcal{T}_r + \widetilde{\mathcal{V}}_r^{\mathrm{int,BM}}.$$

Differentiating along a solution of Equation (109) gives

$$\begin{aligned} \frac{d\widetilde{\mathcal{H}}_r^{\mathrm{BM}}}{dt} &= \dot{\boldsymbol{q}}^T \left[\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}_r^{\mathrm{int,BM}} \right] \\ &= \dot{\boldsymbol{q}}^T \left[\nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}_r^{\mathrm{int,BM}} - \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \mathcal{V}_{e,r}^{\mathrm{int}} \right]. \end{aligned} \tag{112}$$

For this derivative to vanish for arbitrary positions and velocities, the vector in brackets must vanish. Thus preservation of the inherited Lagrangian form and this energy-conservation construction require the same gradient identity.

Applying Equation (79) to $\boldsymbol{F}_{\mathrm{BM}}$, with $\phi_e = \mathcal{V}_{e,r}^{\mathrm{int}}$, gives the condition for this internal vector to derive from one common potential Φ_{BM}:

$$\begin{aligned} &\frac{\partial}{\partial q_j} \left[\sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \mathcal{V}_{e,r}^{\mathrm{int}}}{\partial q_i} \right] - \frac{\partial}{\partial q_i} \left[\sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,j} \frac{\partial \mathcal{V}_{e,r}^{\mathrm{int}}}{\partial q_j} \right] \\ &\quad = \sum_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \mathcal{V}_{e,r}^{\mathrm{int}}}{\partial q_i \, \partial q_j} = 0, \qquad i,j = 1,\ldots,n. \end{aligned} \tag{113}$$

The identity must hold throughout the coordinate domain; agreement at training states alone does not establish it elsewhere.

A nonzero mixed-derivative difference rules out a common potential locally and the associated mechanical Lagrangian and energy-conservation construction with the same reduced kinetic energy. Another variational formulation with a kinetic energy different from $\mathcal{T}_r$ is not excluded.

The same mixed derivatives determine the tangent stiffness. Defining $\boldsymbol{K}_{e,r} = \nabla_{\boldsymbol{q}}^2 \mathcal{V}_{e,r}^{\mathrm{int}}$ and differentiating the two internal terms gives

$$\boldsymbol{K}_r^{\mathrm{ECSW}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{K}_{e,r}, \qquad \boldsymbol{K}_r^{\mathrm{BM}} = \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{K}_{e,r}. \tag{114}$$

Consequently,

$$[\boldsymbol{K}_r^{\mathrm{BM}}]_{ij} - [\boldsymbol{K}_r^{\mathrm{BM}}]_{ji} = \sum_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \mathcal{V}_{e,r}^{\mathrm{int}}}{\partial q_i \, \partial q_j}. \tag{115}$$

The same nonzero difference therefore excludes the required potential and makes the BM tangent stiffness nonsymmetric. Symmetry alone does not establish stability; the following example first examines the potential condition, then the resulting motion.

A.3.1 Two-element example: potential compatibility

Consider two reduced element potentials,

$$\mathcal{V}_{1,r}^{\text{int}} = \frac{1}{2}[q_1 + q_2]^2, \qquad \mathcal{V}_{2,r}^{\text{int}} = \frac{1}{2}[q_1 + 2q_2]^2. \tag{116}$$

Differentiating gives their local internal contributions:

$$\nabla_{\boldsymbol{q}} \mathcal{V}_{1,r}^{\text{int}} = \begin{pmatrix} q_1 + q_2 \\ q_1 + q_2 \end{pmatrix}, \qquad \nabla_{\boldsymbol{q}} \mathcal{V}_{2,r}^{\text{int}} = \begin{pmatrix} q_1 + 2q_2 \\ 2[q_1 + 2q_2] \end{pmatrix}. \tag{117}$$

Both rules select both elements with positive weights. Their weighted internal vectors are compared using the compatibility test in Equation (113).

Classical ECSW

One scalar weight per element gives the sampled potential

$$\begin{aligned} \xi_1 = \xi_2 = 1, \\ \widetilde{\mathcal{V}}_r^{\text{int}} = \mathcal{V}_{1,r}^{\text{int}} + \mathcal{V}_{2,r}^{\text{int}}. \end{aligned} \tag{118a}$$

Assembling the weighted local gradients gives

$$\begin{aligned} \nabla_{\boldsymbol{q}} \widetilde{\mathcal{V}}_r^{\text{int}} &= \sum_{e=1}^{2} \xi_e \nabla_{\boldsymbol{q}} \mathcal{V}_{e,r}^{\text{int}} \\ &= \begin{pmatrix} q_1 + q_2 \\ q_1 + q_2 \end{pmatrix} + \begin{pmatrix} q_1 + 2q_2 \\ 2[q_1 + 2q_2] \end{pmatrix} \\ &= \begin{pmatrix} 2q_1 + 3q_2 \\ 3q_1 + 5q_2 \end{pmatrix}. \end{aligned} \tag{119a}$$

Differentiating the first component with respect to q_2 and the second with respect to q_1 gives:

The crossed derivatives agree:

$$\begin{aligned} \frac{\partial^2 \widetilde{\mathcal{V}}_r^{\text{int}}}{\partial q_2 \, \partial q_1} &= \frac{\partial [2q_1 + 3q_2]}{\partial q_2} = 3, \\ \frac{\partial^2 \widetilde{\mathcal{V}}_r^{\text{int}}}{\partial q_1 \, \partial q_2} &= \frac{\partial [3q_1 + 5q_2]}{\partial q_1} = 3. \end{aligned} \tag{120a}$$

Equation (113) gives these differences directly from the element contributions:

Each element shares its weight between the two components:

$$\begin{aligned} &\sum_{e=1}^{2} [\xi_e - \xi_e] \frac{\partial^2 \mathcal{V}_{e,r}^{\text{int}}}{\partial q_1 \, \partial q_2} \\ &\quad = [1 - 1] \cdot 1 + [1 - 1] \cdot 2 \\ &\quad = 0. \end{aligned} \tag{121a}$$

BM-ECSW

The diagonal matrices specify the weights for each reduced component:

$$\begin{aligned} \boldsymbol{D}_1 = \text{diag}(1, 4), \\ \boldsymbol{D}_2 = \boldsymbol{I}_2. \end{aligned} \tag{118b}$$

Assembling the weighted local gradients gives

$$\begin{aligned} &\sum_{e=1}^{2} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \mathcal{V}_{e,r}^{\text{int}} \\ &\quad = \begin{pmatrix} q_1 + q_2 \\ 4[q_1 + q_2] \end{pmatrix} + \begin{pmatrix} q_1 + 2q_2 \\ 2[q_1 + 2q_2] \end{pmatrix} \\ &\quad = \begin{pmatrix} 2q_1 + 3q_2 \\ 6q_1 + 8q_2 \end{pmatrix}. \end{aligned} \tag{119b}$$

The crossed derivatives differ:

$$\begin{aligned} \frac{\partial [2q_1 + 3q_2]}{\partial q_2} &= 3, \\ \frac{\partial [6q_1 + 8q_2]}{\partial q_1} &= 6. \end{aligned} \tag{120b}$$

Element 1 has different weights in the two components:

$$\begin{aligned} &\sum_{e=1}^{2} [\xi_{e,1} - \xi_{e,2}] \frac{\partial^2 \mathcal{V}_{e,r}^{\text{int}}}{\partial q_1 \, \partial q_2} \\ &\quad = [1 - 4] \cdot 1 + [1 - 1] \cdot 2 \\ &\quad = -3. \end{aligned} \tag{121b}$$

The sampled potential generates both components of the classical internal vector. It defines the Lagrangian in Equation (106), and the conserved-energy construction in Equation (108) applies.

Element 1 contributes -3 and element 2 contributes zero, so the differences do not cancel. No smooth scalar potential can generate this BM internal vector. A Lagrangian of the form in Equation (110) therefore does not exist, and the original kinetic energy plus a potential depending only on $\boldsymbol{q}$ cannot be conserved for all motions.

A.3.2 Stiffness and stability of the two-element example

To examine stability, differentiate the internal contributions once more. The element Hessians are positive semidefinite:

$$\boldsymbol{K}_{1,r} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{K}_{2,r} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Their classical and BM weighted sums give

$$\boldsymbol{K}_r^{\mathrm{ECSW}} = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, \qquad \boldsymbol{K}_r^{\mathrm{BM}} = \begin{pmatrix} 2 & 3 \\ 6 & 8 \end{pmatrix}. \tag{122}$$

These stiffness matrices are exact Jacobians of the internal vectors, whereas Section A.2.3 examines approximate correction matrices. Their off-diagonal entries are the mixed derivatives computed above. The classical stiffness is positive definite because, for every nonzero $\boldsymbol{z} = (z_1, z_2)^T$,

$$\boldsymbol{z}^T \boldsymbol{K}_r^{\mathrm{ECSW}} \boldsymbol{z} = [z_1 + z_2]^2 + [z_1 + 2z_2]^2 > 0;$$

both squares vanish only when $\boldsymbol{z} = \boldsymbol{0}$.

The BM stiffness has determinant $2 \cdot 8 - 3 \cdot 6 = -2$ and a negative eigenvalue $\lambda_- = 5 - 3\sqrt{3}$. With $\boldsymbol{M}_r = \boldsymbol{I}_2$, motion along the associated eigenvector satisfies $\ddot{a} + \lambda_- a = 0$ and admits exponential growth proportional to $\exp(\sqrt{-\lambda_-}\, t)$.

The structural counterexample shows that positive modal weights can introduce an unstable mode into a stable classical model. A training fit may exclude these particular weights, but nonnegativity alone does not. Whether a flow HPROM exhibits a similar deterioration must be examined through its own online equations and trajectories.

A.4 Computational evidence: gradient structure and online behavior

The computations examine three questions: whether the projected vector retains a common scalar objective (Section A.1), how the correction matrix behaves (Section A.2.3), and whether the online trajectory remains accurate. The Burgers tests address all three; AERO-F provides further evidence of online failure. These flow computations do not test the mechanical conservation properties of Section A.3.

A.4.1 Burgers: gradient structure and online behavior

The classical and BM rules use the same 96-mode ROB, with $N_c = 90$ or 225. Their trajectories use the online settings of Section 4.2.

The common-objective test uses the exact Jacobian $\boldsymbol{J}_n = \partial \widetilde{\boldsymbol{r}}_n / \partial \boldsymbol{q}$, with the previous state held fixed. If the projected vector derives from one common scalar objective, its exact Jacobian must be symmetric by Equation (74). For example, at the first-step predictor with $\boldsymbol{\mu} = (4.56, 0.019)$, two crossed entries of the exact Jacobian for BM with $N_c = 225$ are

$$[\boldsymbol{J}_n]_{12} = \frac{\partial \widetilde{r}^{\mathrm{BM}}_{n,1}}{\partial q_2} = -0.766, \qquad [\boldsymbol{J}_n]_{21} = \frac{\partial \widetilde{r}^{\mathrm{BM}}_{n,2}}{\partial q_1} = +0.124. \tag{123}$$

This unequal pair already shows that the Jacobian is nonsymmetric at this initial guess. Table 4 applies the same test to all crossed entries together, at the computed states of each HPROM for the three test parameter points of Section 4.2. With 500 time steps per parameter, this gives $3 \times 500 = 1\,500$ exact-Jacobian evaluations per HPROM. The reported ratio $\|\boldsymbol{J}_n - \boldsymbol{J}_n^T\|_F / \|\boldsymbol{J}_n\|_F$ is zero when all crossed entries agree and positive when any pair differs.

Table 4
Burgers: exact-Jacobian gradient test. Ratios are median relative asymmetries; nonsymmetric counts exclude roundoff.

Model	N_c	$\|\boldsymbol{J}_n - \boldsymbol{J}_n^T\|_F / \|\boldsymbol{J}_n\|_F$	Nonsymmetric
HPROM (ECSW)	90	8.95×10^{-17}	0/1 500
HPROM (ECSW)	225	8.74×10^{-17}	0/1 500
HPROM (BM-ECSW)	90	1.099	1 500/1 500
HPROM (BM-ECSW)	225	0.434	1 500/1 500

Classical ECSW is symmetric to roundoff, whereas both BM rules are nonsymmetric at every evaluated configuration. The unequal derivatives above exclude a common smooth objective locally for the accurate $N_c = 225$ rule. Loss of gradient structure therefore does not, by itself, distinguish the successful rule from the unsuccessful one.

The correction matrices require the separate check of Section A.2.3. Table 5 reports the median ratio $\lambda_{\min}/\lambda_{\max}$ for $\mathrm{sym}(\widehat{\boldsymbol{J}}_n)$, evaluated at the first nonlinear iterate of each of the 1 500 time steps. Here $\lambda_{\max} > 0$, so positive and negative ratios indicate positive definiteness and indefiniteness, respectively; the indefinite count records occurrences of negative directions. The maximum trajectory error is taken over the three parameters using Equation (71), and is distinct from the training-fit error.

Table 5
Burgers: sampled meshes, correction-matrix diagnostics, and maximum trajectory errors.

Model	N_c	n_e	Median $\lambda_{\min}/\lambda_{\max}$	Indefinite	$\mathbb{RE}^{\max}_{2,\boldsymbol{u}}$ [%]
HPROM (ECSW)	90	1 590	+0.678	0/1 500	1.096
HPROM (ECSW)	225	2 492	+0.678	0/1 500	1.096
HPROM (BM-ECSW)	90	145	−0.231	1 500/1 500	4 411.533
HPROM (BM-ECSW)	225	300	+0.046	0/1 500	1.102

The classical rules remain accurate at both values of N_c. BM with $N_c = 225$ reaches comparable accuracy on a much smaller mesh; BM with $N_c = 90$ fails despite a training error of 1.28×10^{-15}. The latter attains the online tolerance at only 13 of 1 500 steps, compared with every step for the $N_c = 225$ rule. Figure 4 shows the accompanying trajectory error and symmetric-part diagnostics.

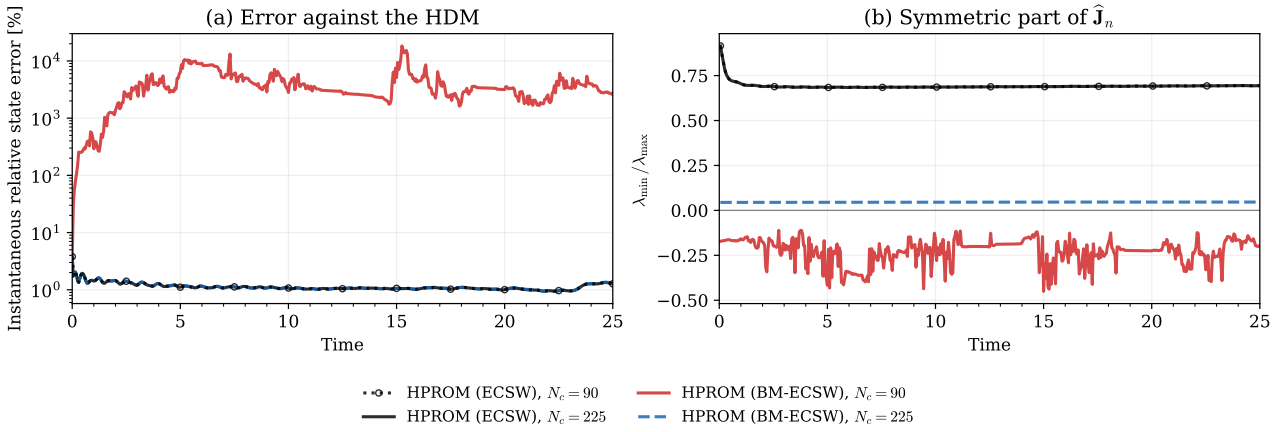


Figure 4: Burgers at $\boldsymbol{\mu} = (4.56, 0.019)$: (a) instantaneous relative state error against the HDM; (b) symmetric-part eigenvalue ratio at the first nonlinear iterate. The classical ECSW curves nearly overlap.

Accuracy of individual steps. This test checks whether BM can produce an inaccurate next state even when its equations are solved accurately. A reference PROM without hyperreduction is computed for 20 time steps at $\boldsymbol{\mu} = (4.56, 0.019)$, using the same 96-mode ROB and evaluating the residual over the full mesh.

At steps 1, 5, and 20, the PROM, both classical rules, and both BM rules receive the same previous state $\boldsymbol{q}^m$ from this reference trajectory. Each model then computes its next state $\boldsymbol{q}^{m+1}$ by solving its own projected equations to a relative residual below 10^{-11}. Comparing these states measures the error introduced in one step, before it accumulates along a trajectory.

At the first step, Table 6 shows that BM with $N_c = 90$ differs from the PROM by 30.63%, whereas the $N_c = 225$ rule differs by 0.048% and classical ECSW nearly reproduces the PROM state. Thus BM can satisfy its equations accurately and still produce an inaccurate state. Failure of the nonlinear solver alone cannot explain the unsuccessful trajectory.

Table 6
Burgers first-step accuracy at $\boldsymbol{\mu} = (4.56, 0.019)$. State errors are against the PROM. The symmetric-part ratio and condition number κ_2 refer to the correction matrix at the shared initial guess.

Model	$\lambda_{\min}/\lambda_{\max}$	$\kappa_2(\widehat{\boldsymbol{J}}_n)$	State error [%]
PROM	+0.915	1.09	0
HPROM (ECSW), $N_c = 90$	+0.915	1.09	4.06×10^{-4}
HPROM (ECSW), $N_c = 225$	+0.915	1.09	9.17×10^{-6}
HPROM (BM-ECSW), $N_c = 90$	−0.172	3 233.39	30.626
HPROM (BM-ECSW), $N_c = 225$	+0.044	13.40	0.048

A.4.2 AERO-F: divergence and limits of matrix diagnostics

The AERO-F example considers two-dimensional unsteady laminar flow past a circular cylinder. The compressible Navier–Stokes equations are discretized by a finite-volume method at a freestream Mach number of 0.2. Linear LSPG models use POD bases constructed from HDM solution snapshots, with classical ECSW or BM-ECSW for hyperreduction.

The cylinder computations connect to Section A.2.4: a BM HPROM can diverge while every correction-matrix eigenvalue retains a positive real part. Unlike the failed Burgers rule, whose correction matrices also have eigenvalues with negative real parts, this case shows why positive real parts alone do not establish reliable online behavior.

Table 7 compares three BM rules, all evaluated at Re = 100. The nonparametric model is trained at this Reynolds number, with $n = 35$ and 376 training snapshots. The Reynolds-parametric model uses a common basis with $n = 96$, constructed from HDM trajectories at seven Reynolds numbers between 70 and 180; its sampling rules use 385 training snapshots. All rules attain their training tolerance. The symmetric-part ratio is the median over the online corrections, including repeated-step corrections, with the same sign interpretation as in Table 5.

Table 7
AERO-F at Re = 100: BM correction matrices and online outcomes. Counts include repeated-step corrections: 226 matrices over 215 steps for the diverging run.

Model	n	τ	n_e	$\frac{\lVert \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} - [\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}]^T \rVert_F}{\lVert \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} \rVert_F}$	$\frac{\lambda_{\min}}{\lambda_{\max}}$	Indefinite	Online solution
Nonparametric	35	10^{-2}	64	0.44	$+6.9 \times 10^{-2}$	0 of 1 500	stable
Parametric	96	10^{-2}	159	0.50	-1.9×10^{-2}	210 of 226	diverges at $t = 21$
Parametric	96	10^{-3}	308	0.33	$+1.6 \times 10^{-1}$	0 of 1 500	stable

For the 159-node rule, the symmetric part is indefinite from the first step. The lift departs from the HDM near $t = 12$, the coordinate norm grows to about 40 times that of the stable HPROMs, and the run stops at $t = 21$ with negative pressures. Figure 5 compares this trajectory with the accurate 308-node BM rule and the classical rule on 837 sampled nodes.

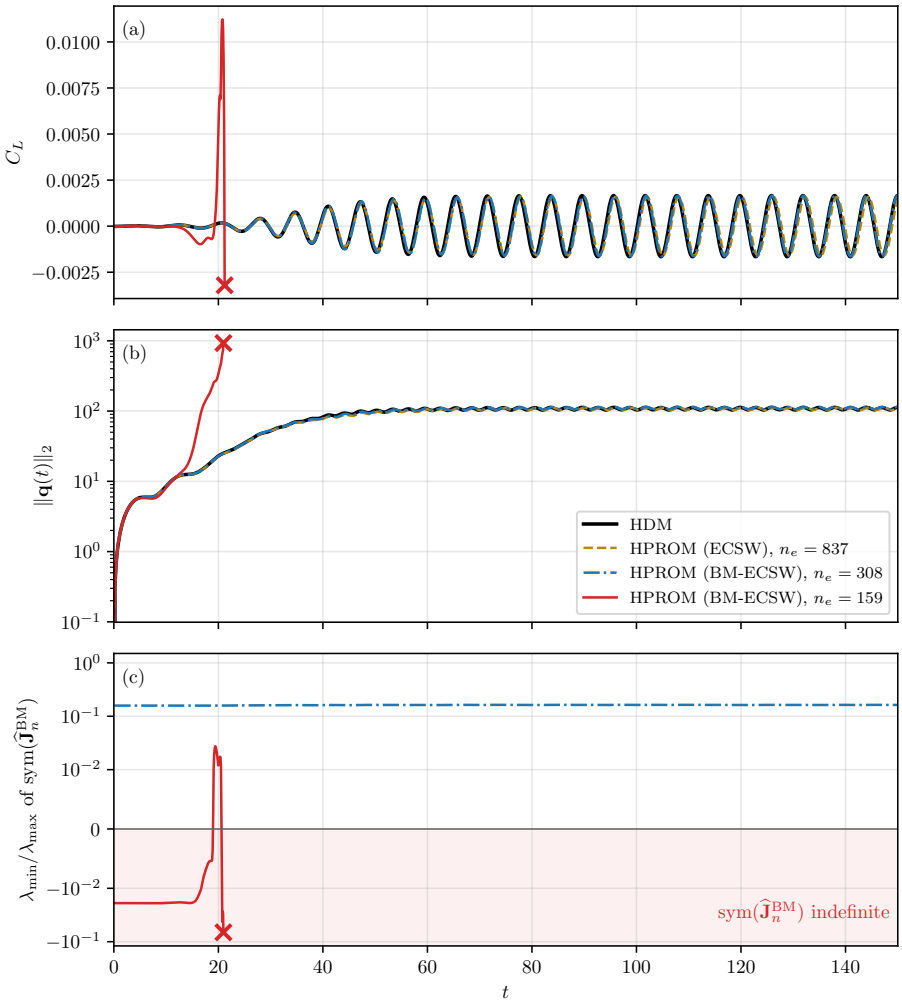


Figure 5: AERO-F at Re = 100 with $n = 96$: (a) lift coefficient; (b) coordinate norm, with the HDM projected onto the ROB; (c) symmetric-part eigenvalue ratio at each BM correction. A cross marks failure.

The sign of the symmetric part is not a complete stability criterion. A 174-node BM rule at $\tau = 10^{-2}$ diverges at $t = 41.5$, although its symmetric part is positive definite from $t = 25$ until collapse. Classical rules restricted to 159 and 174 nodes also diverge, at $t = 56$ and 80: a sufficiently small classical mesh can fail as well.

Conclusion. The successful Burgers BM rule substantially reduces the sampled mesh while retaining the observed trajectory accuracy. Both successful and unsuccessful BM rules lack a common objective at the evaluated configurations, and neither an accurate training fit nor favorable individual matrix diagnostics establish reliable online behavior. The computations establish losses of gradient structure and observed online failures, but do not isolate which loss causes each failure.

B Individual online measurements

Table 8 reports the individual online times in seconds. SD is the sample standard deviation, computed with denominator $3 - 1$.

Table 8
Individual online wall-clock times (seconds) for HPROM (ECSW) and HPROM (BM-ECSW) at the three unsampled parameter points. Each measurement uses a fresh process and 20 CPU threads.

Parameter	Computational model	Run 1	Run 2	Run 3	Mean	SD
$(4.56, 0.019)$	HPROM (ECSW)	18.889	18.945	17.795	18.543	0.648
$(4.56, 0.019)$	HPROM (BM-ECSW)	1.238	1.205	1.212	1.218	0.017
$(4.75, 0.020)$	HPROM (ECSW)	19.682	18.408	18.390	18.827	0.741
$(4.75, 0.020)$	HPROM (BM-ECSW)	1.209	1.204	1.200	1.204	0.005
$(5.19, 0.026)$	HPROM (ECSW)	17.304	19.880	18.526	18.570	1.288
$(5.19, 0.026)$	HPROM (BM-ECSW)	1.206	1.211	1.208	1.208	0.002

Table 9
Fresh production HDM integration times: one run at each queried parameter, 20 CPU threads, and 500 time steps. The last row is the arithmetic mean over the three parameters.

Parameter	Wall-clock time (s)	Wall-clock time (min)
$(4.56, 0.019)$	323.195	5.387
$(4.75, 0.020)$	323.275	5.388
$(5.19, 0.026)$	326.310	5.438
Mean	324.260	5.404

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